

Static analysis and software verification

Lecture 2 - Mathematical background

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- Set $\approx$ Logics
 - subset $\approx$ implication
 - union $\approx$ disjunction (or)
 - intersection $\approx$ conjunction (and)

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- Example: $\leq$ in $\mathbb{Z}$ (informally)
 - Any integer is equal to itself, therefore it is less or equal
 - If an integer i_1 is less or equal than i_2 , and i_2 is less or equal than i_1 , then i_1 and i_2 are the same
 - i_1 is less or equal than i_2 , i_2 is less or equal than i_3 , then i_1 is less or equal than i_3

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 - $|\wp(\{0,1\})| = 4$

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- A set X equipped with a partial order $\subseteq$ is a poset, denoted with $\langle X, \subseteq \rangle$

Exercise

$\supseteq$ on $\wp(X)$ is a partial order?

Exercise

The inverse of a partial order is a partial order?

Exercise

Other examples of posets?

Hasse diagrams

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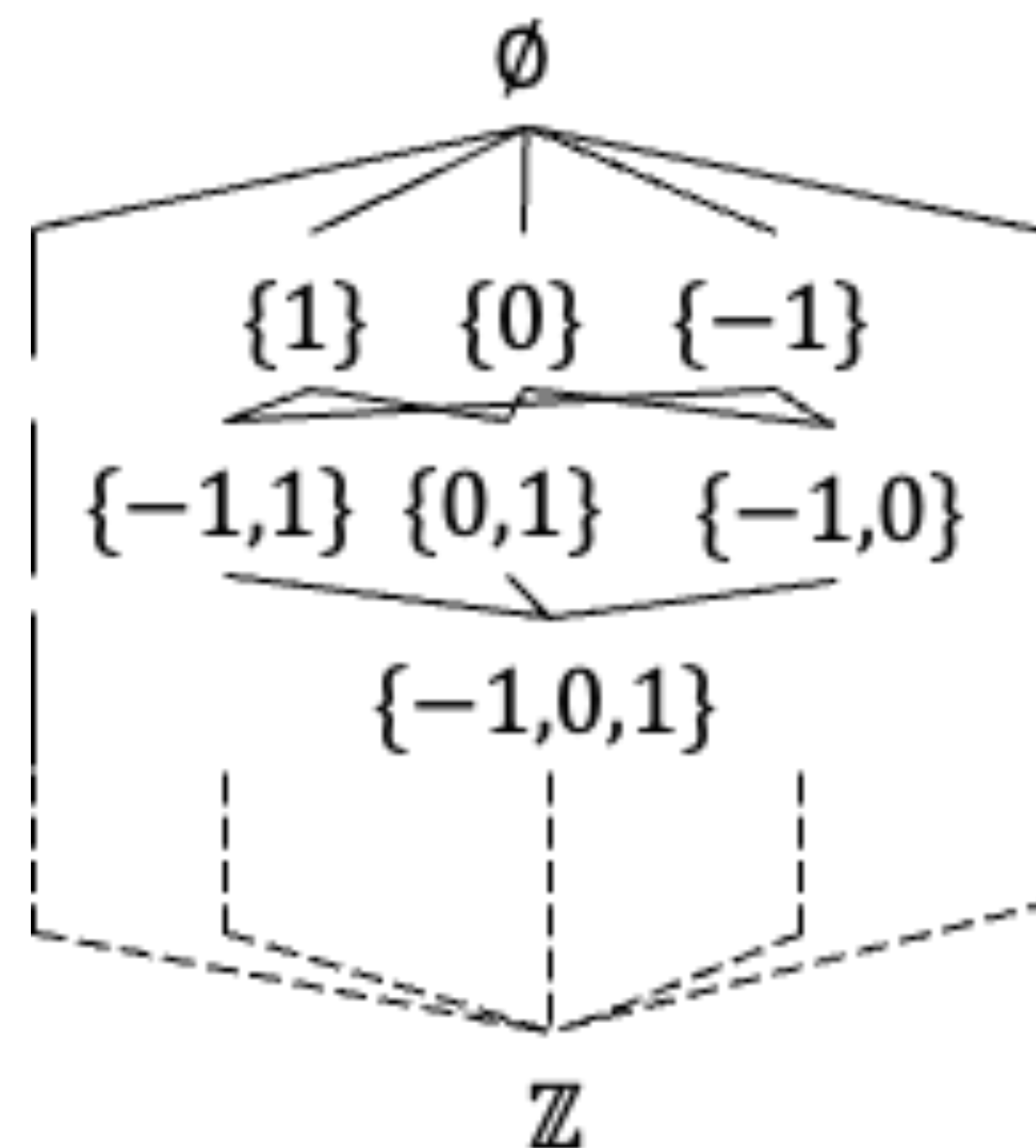
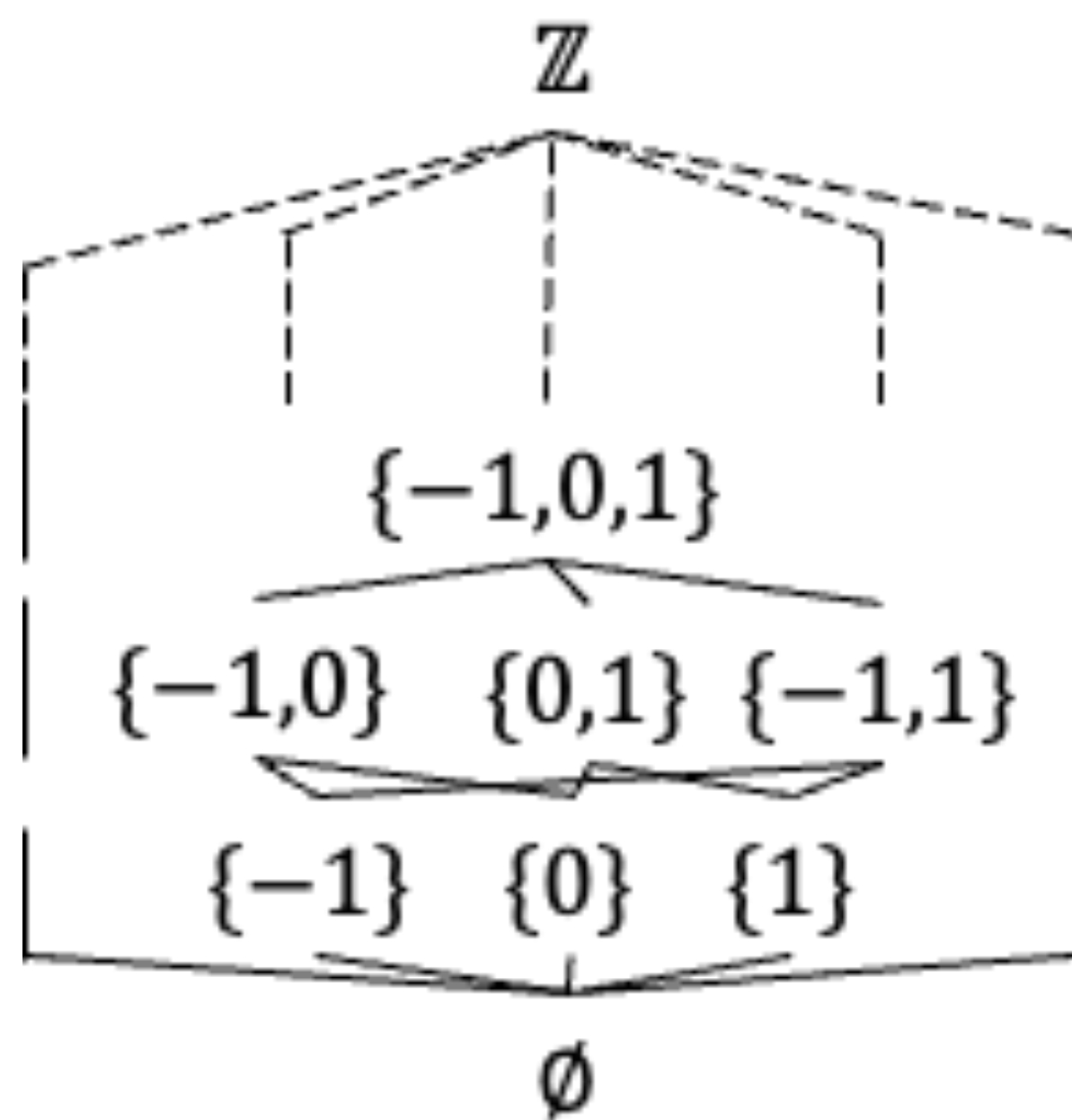
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- Inverse poset: 180° rotation

Hasse diagrams

Examples



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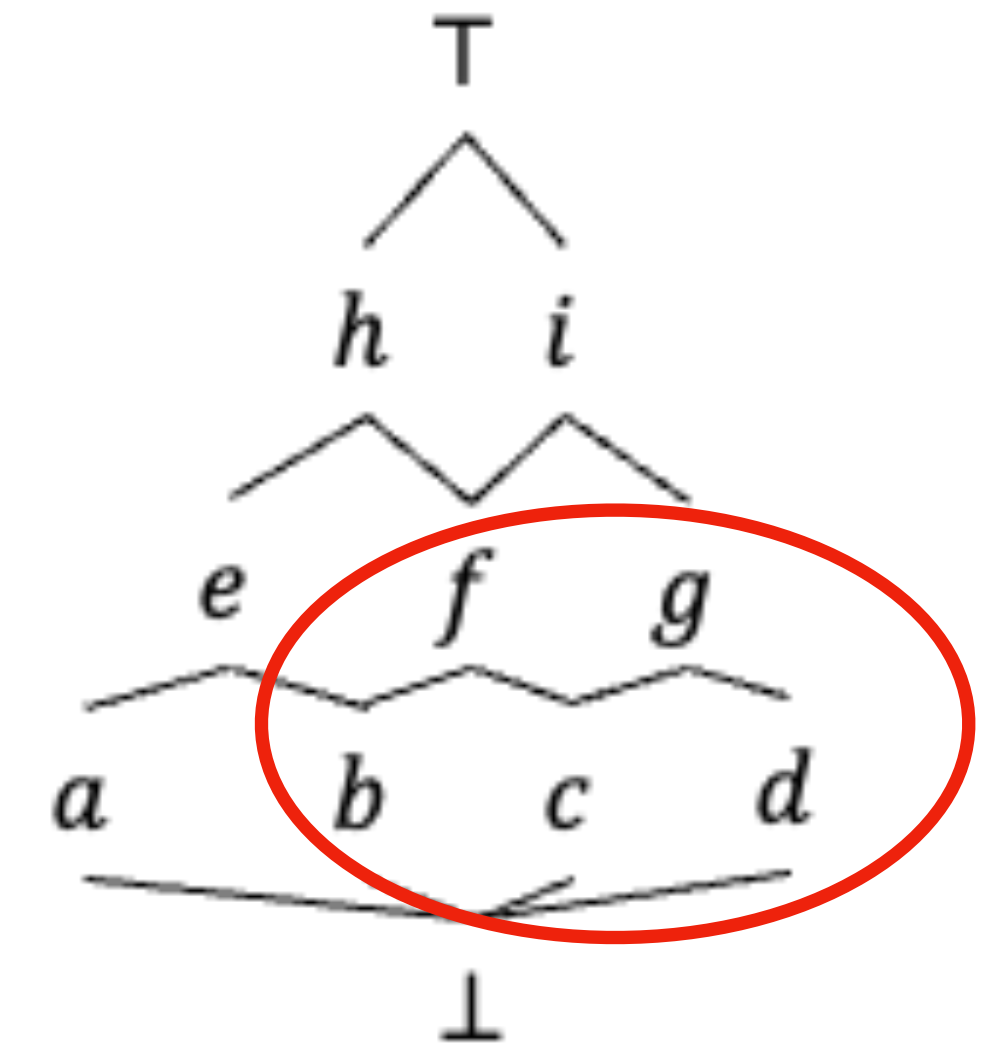
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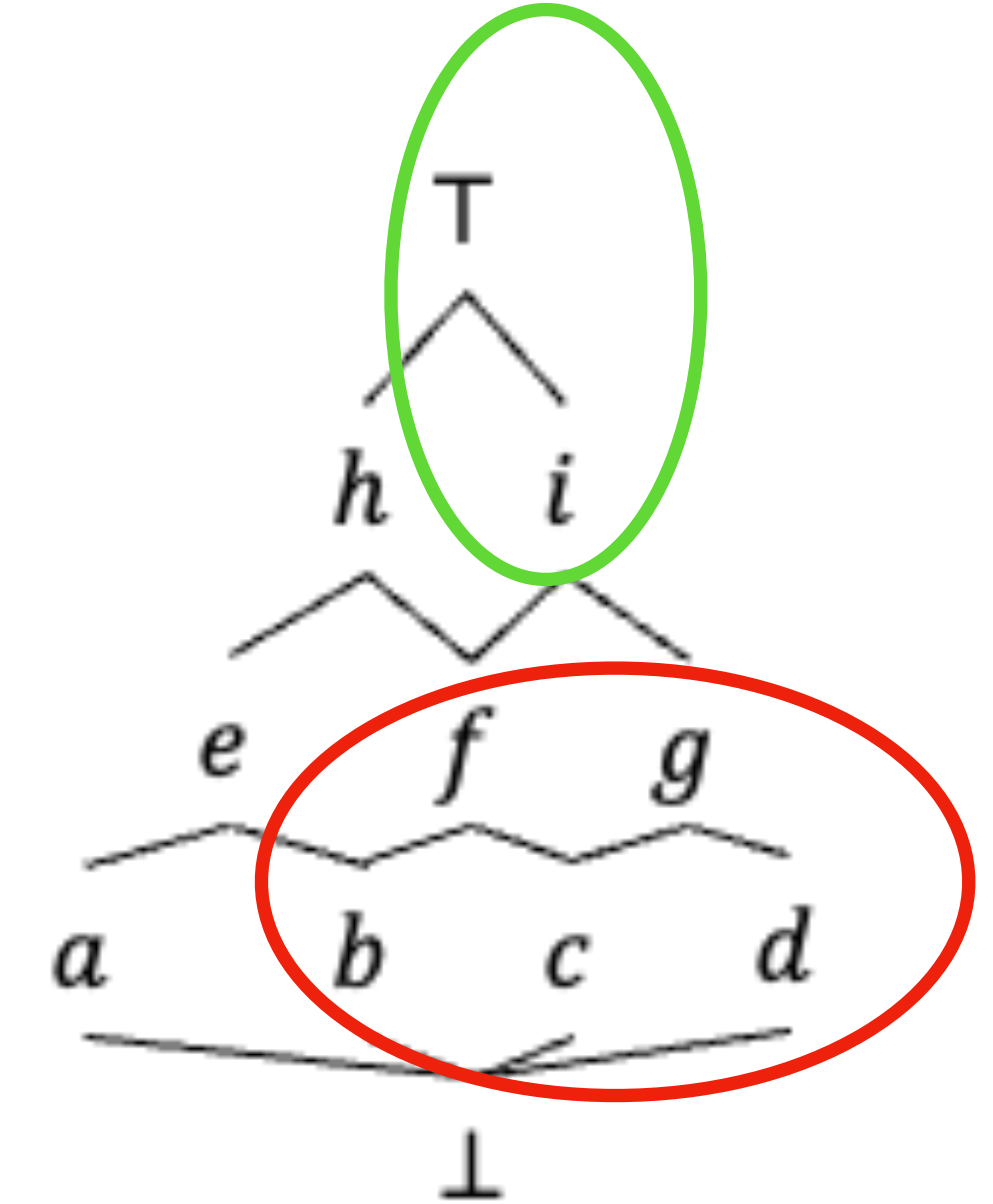
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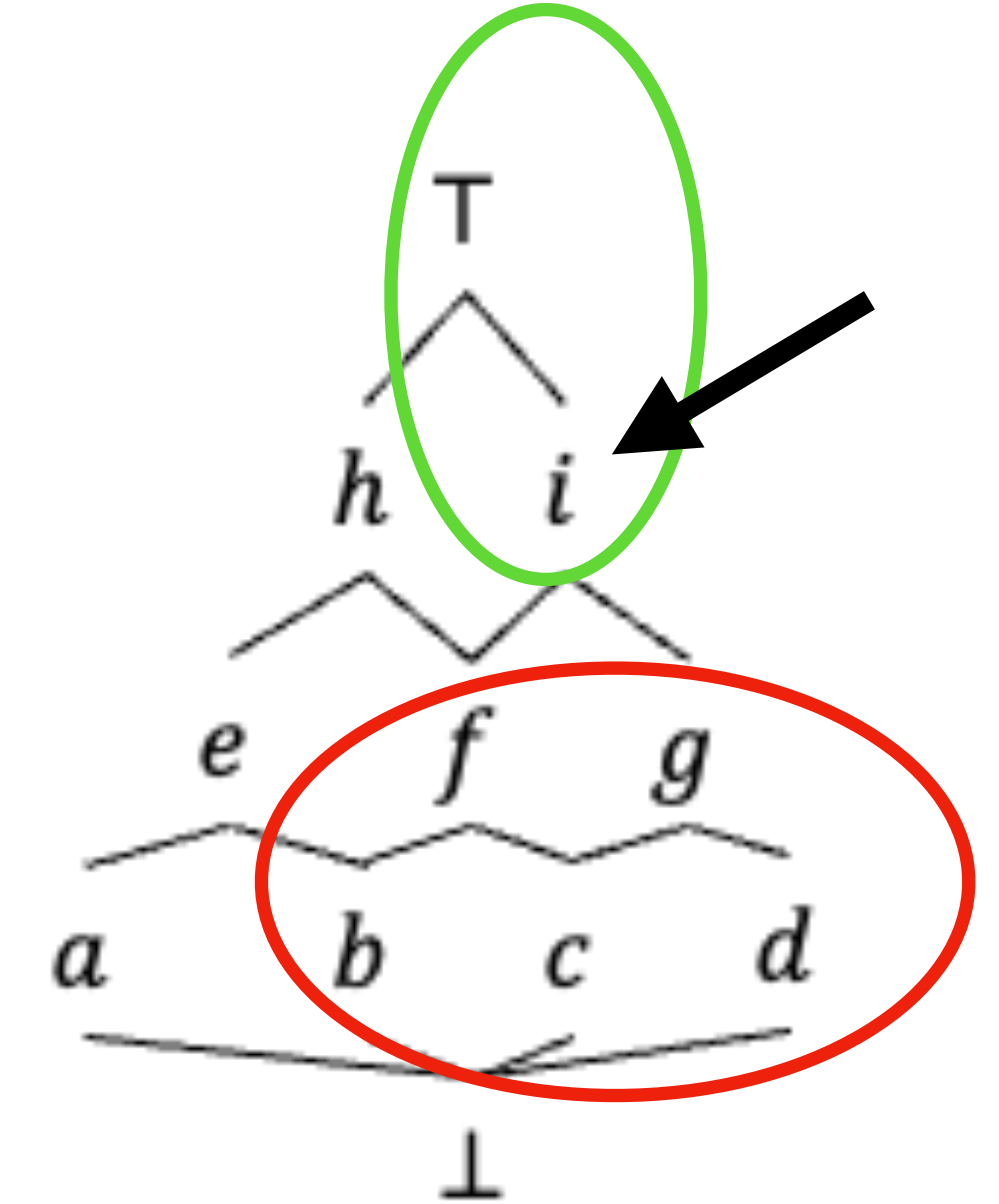
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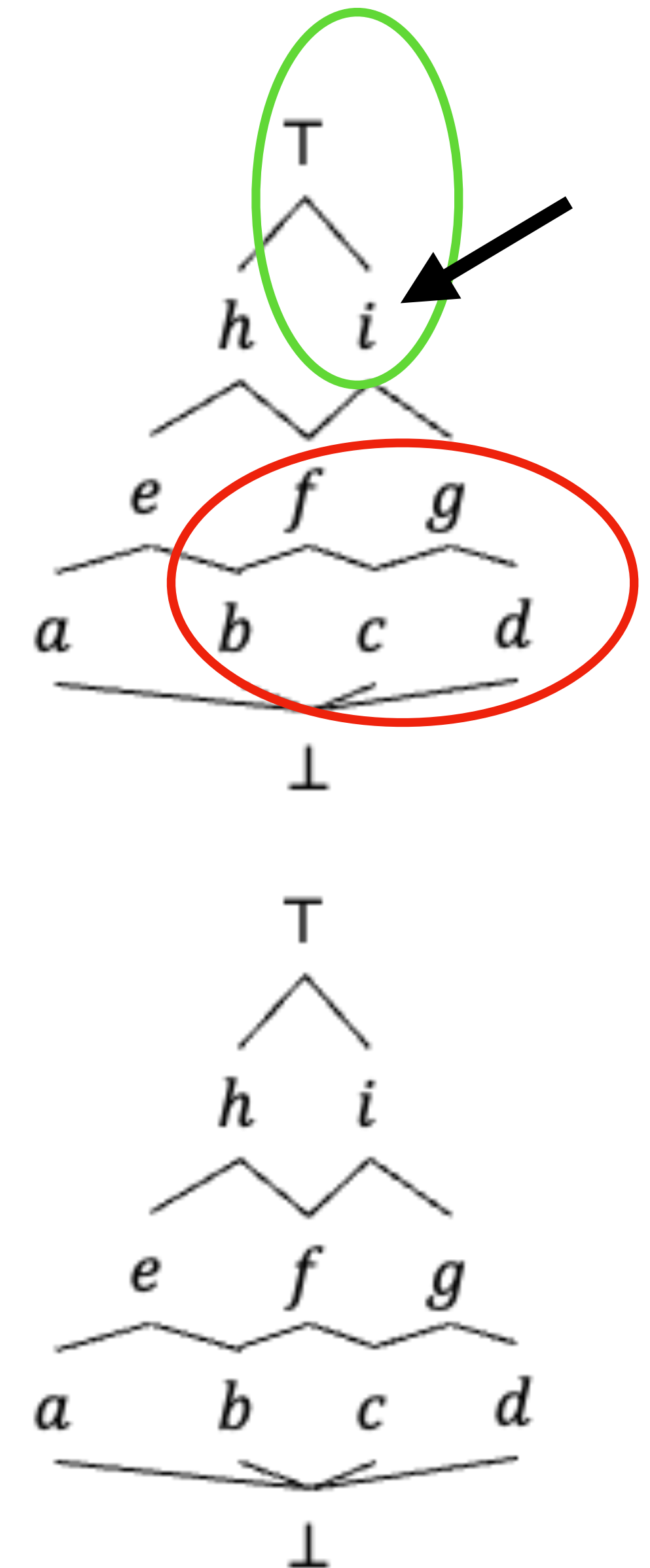
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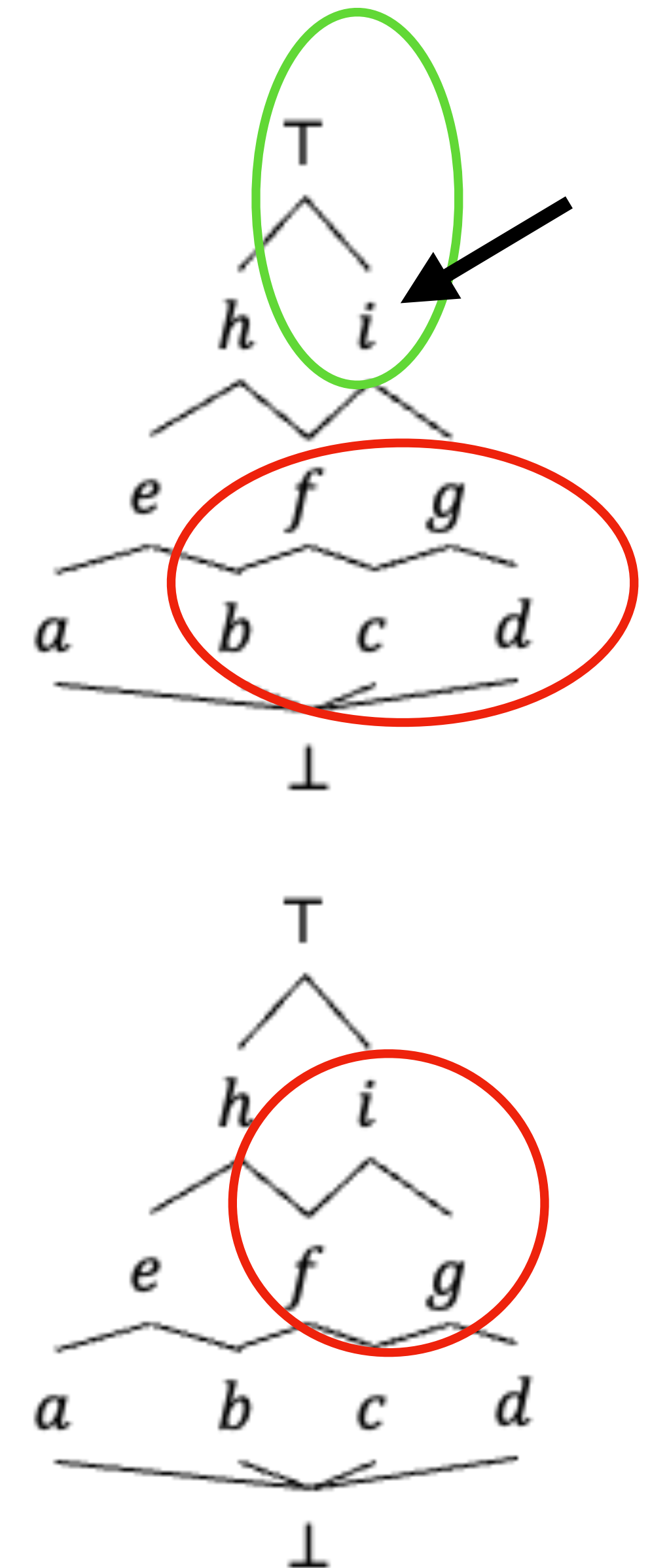
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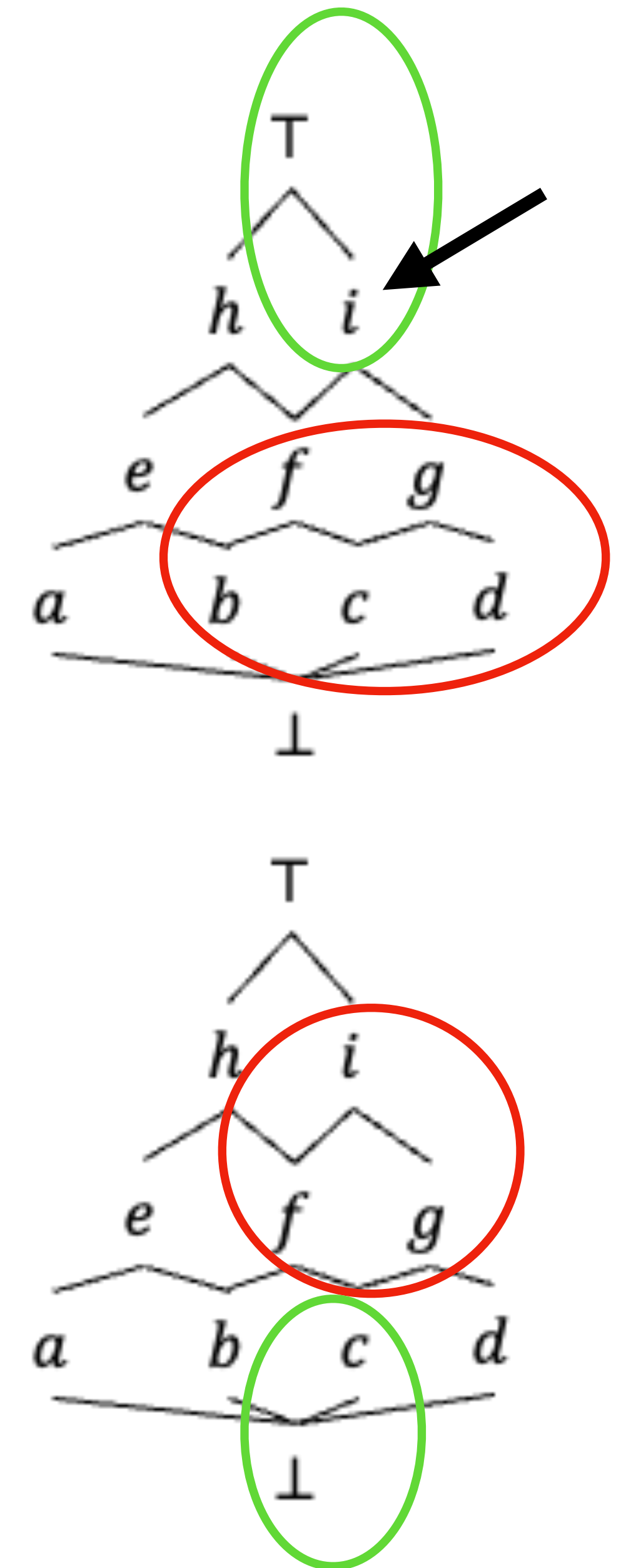
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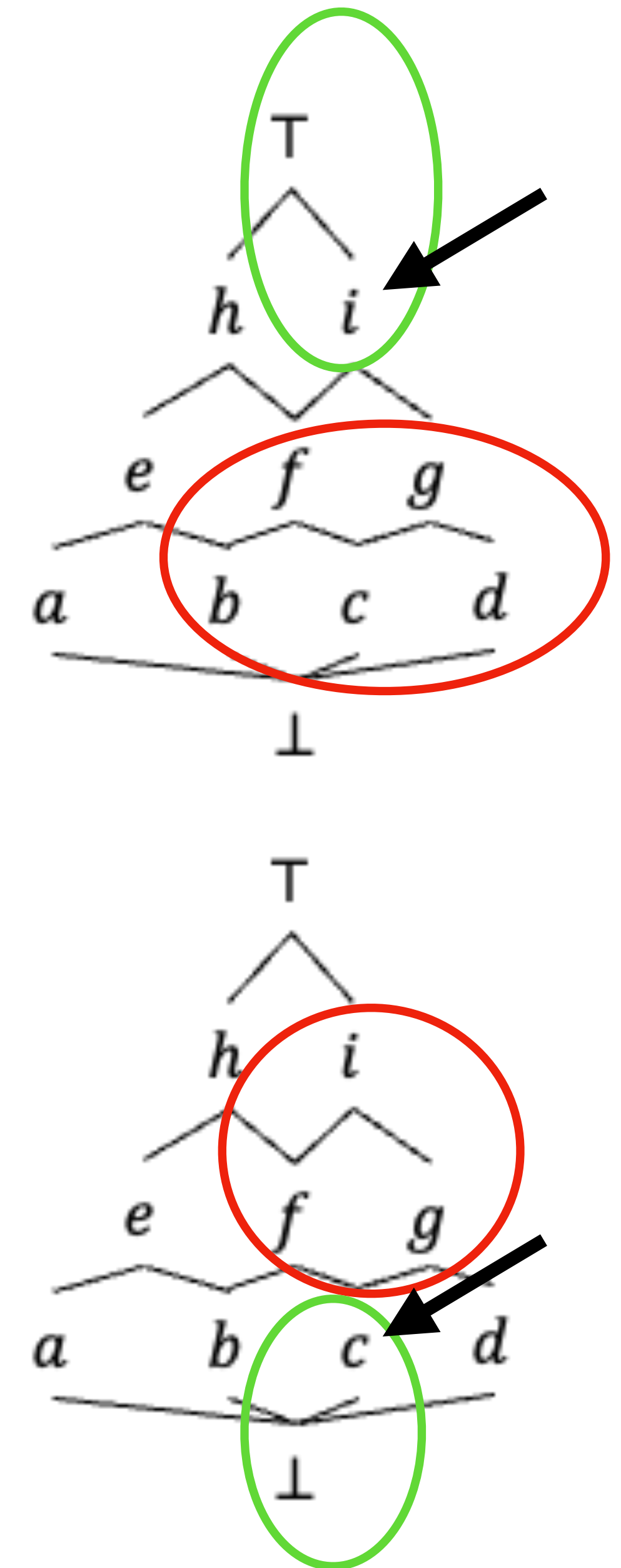
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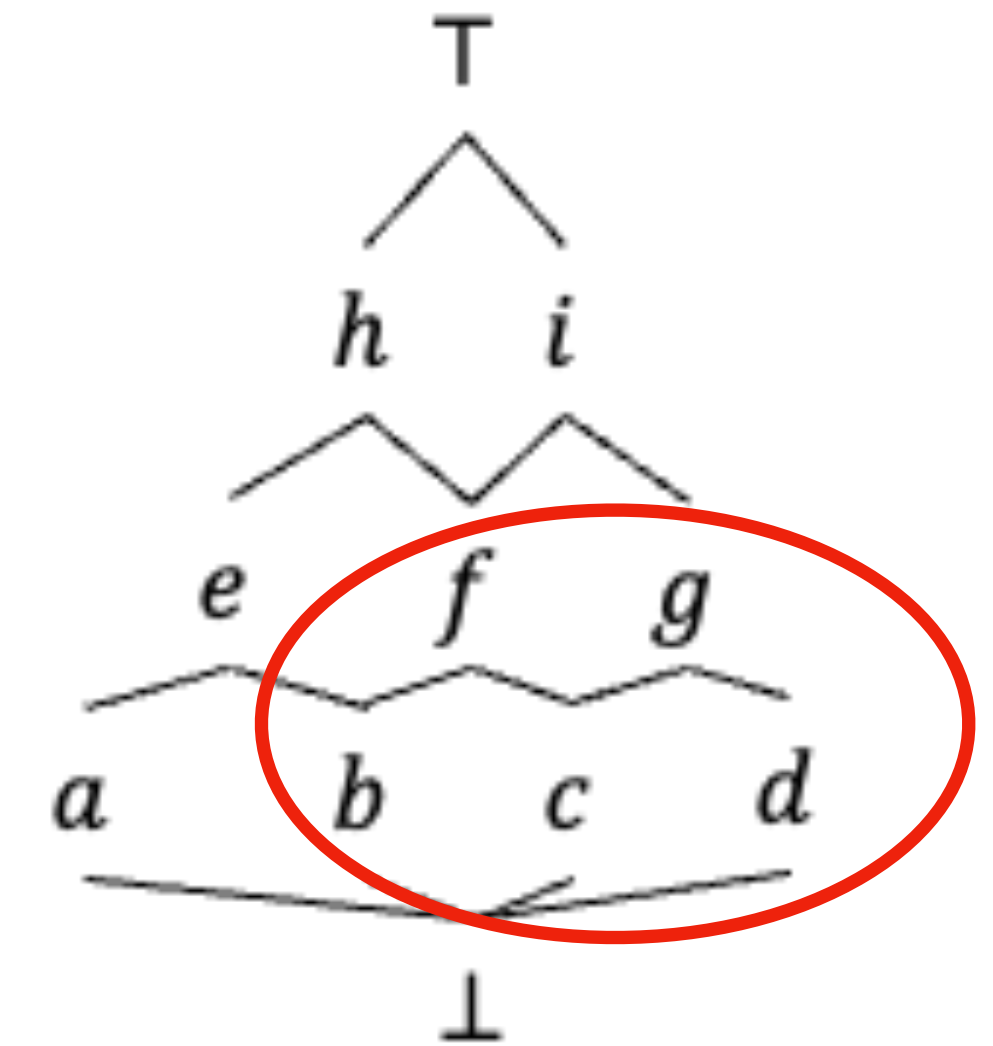
- Given $\langle \wp(X), \subseteq \rangle$, let $S_1, S_2 \in \wp(X)$. Is $S_1 \cup S_2$ the lub of $\{S_1, S_2\}$?
- Given $\langle \wp(X), \subseteq \rangle$, let $S_1, S_2 \in \wp(X)$. Is $S_1 \cap S_2$ the glb of $\{S_1, S_2\}$?

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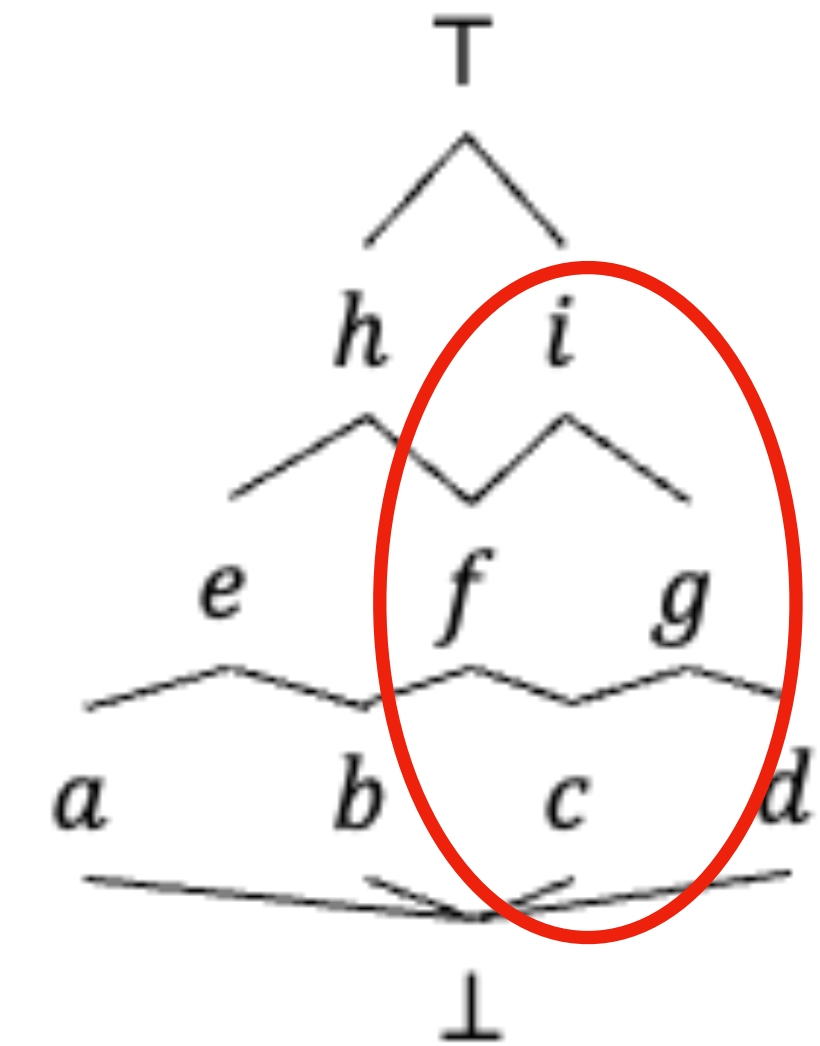
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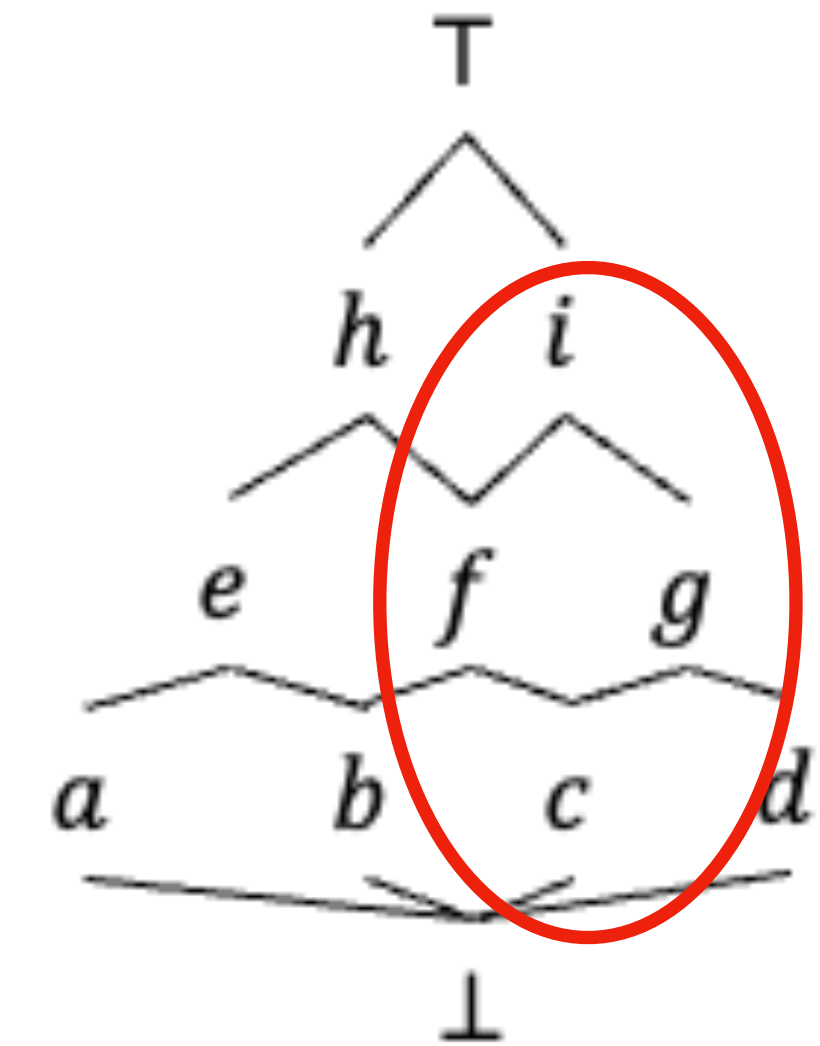
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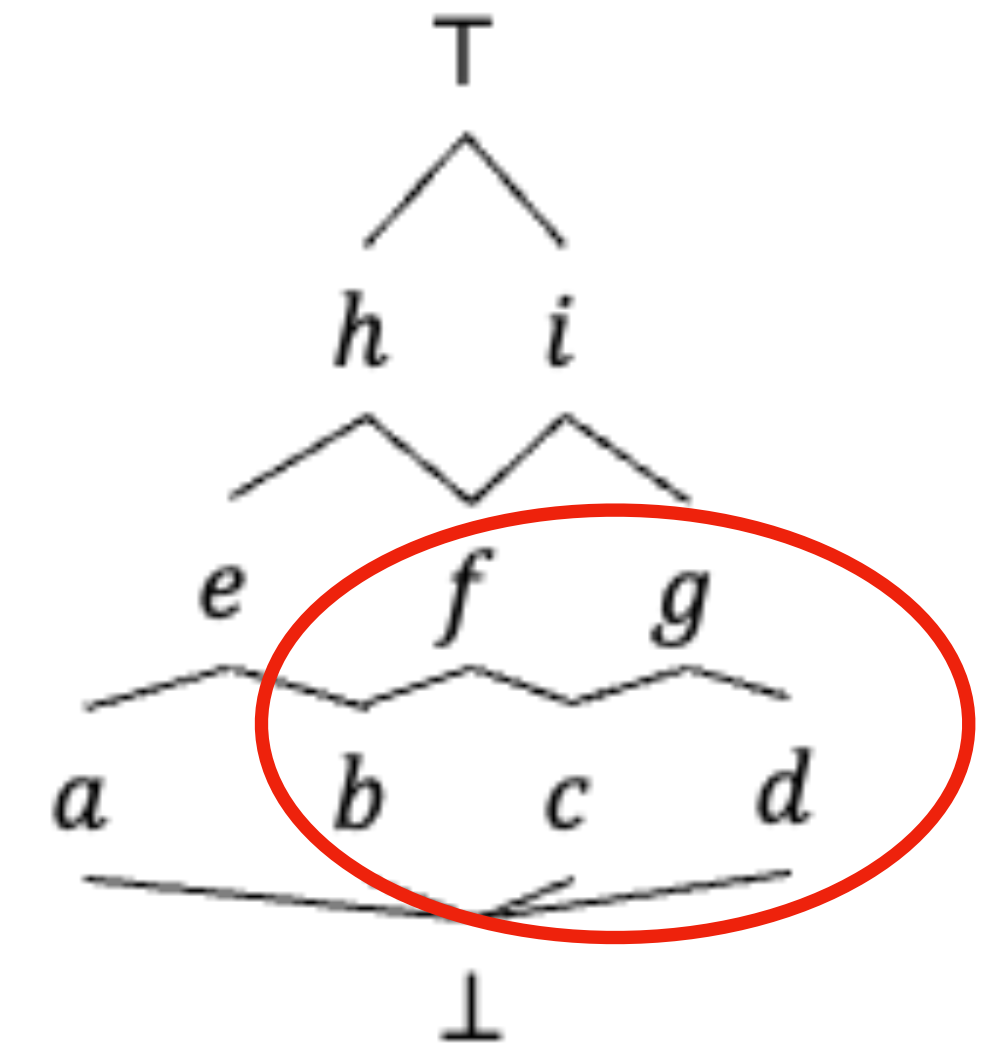
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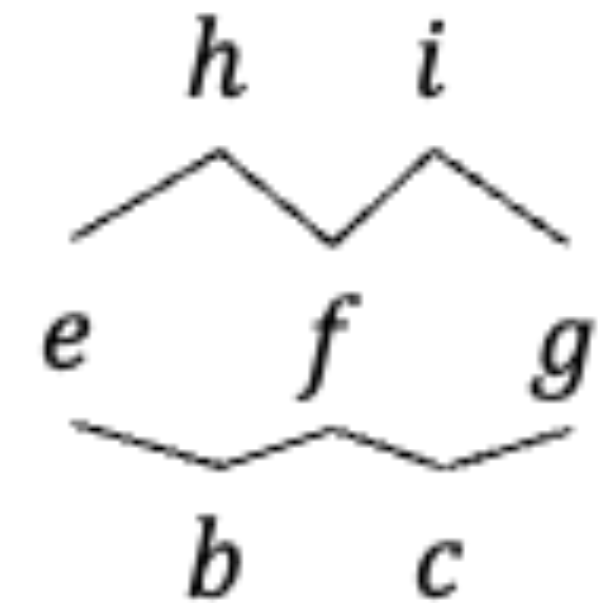
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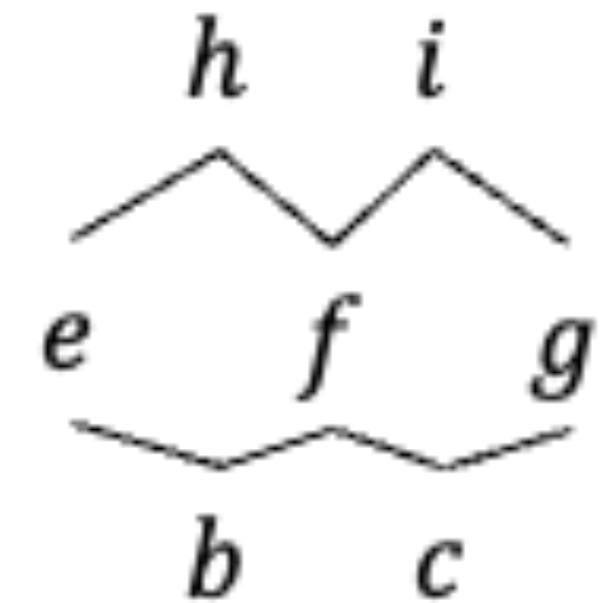
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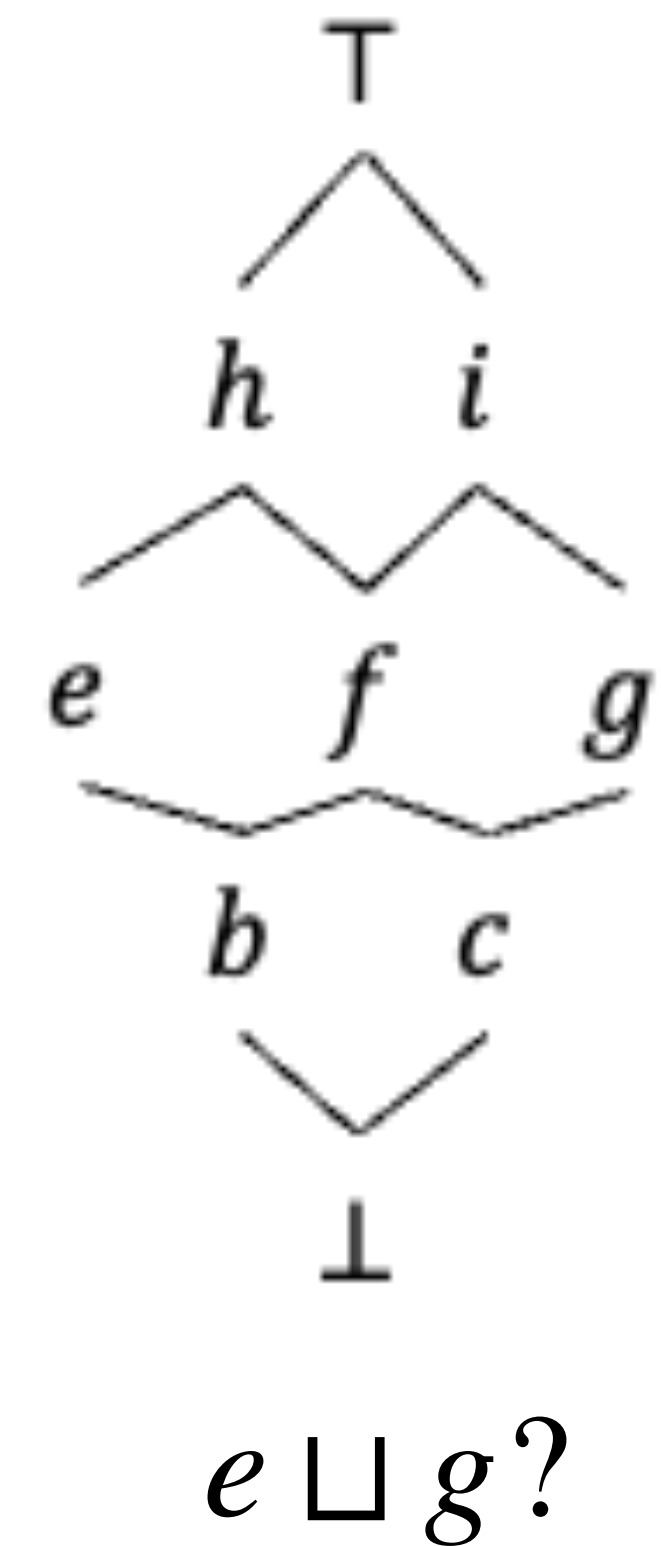
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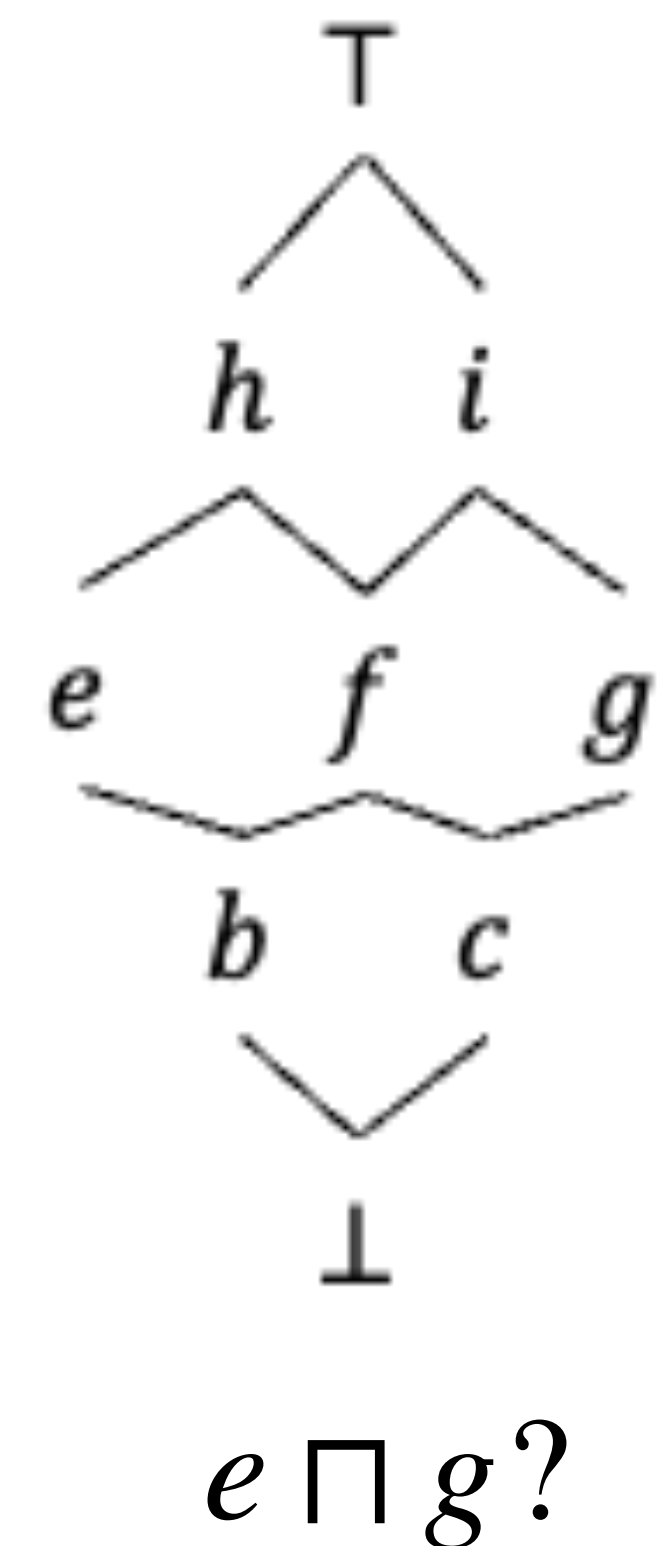
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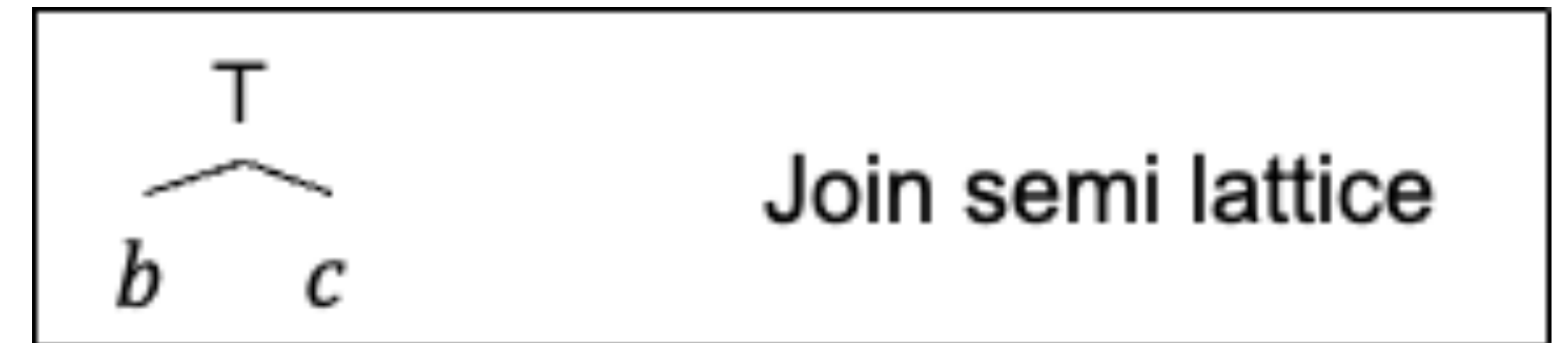
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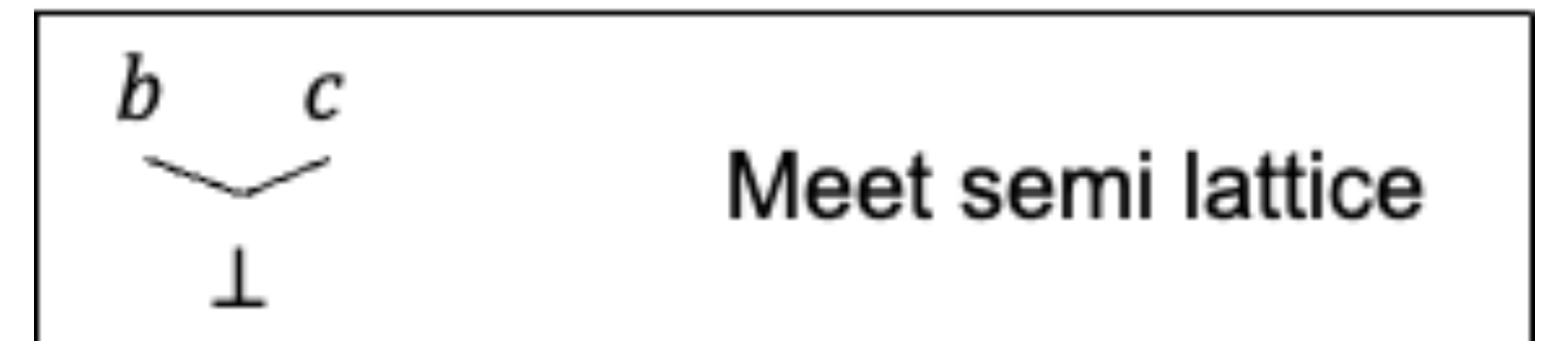
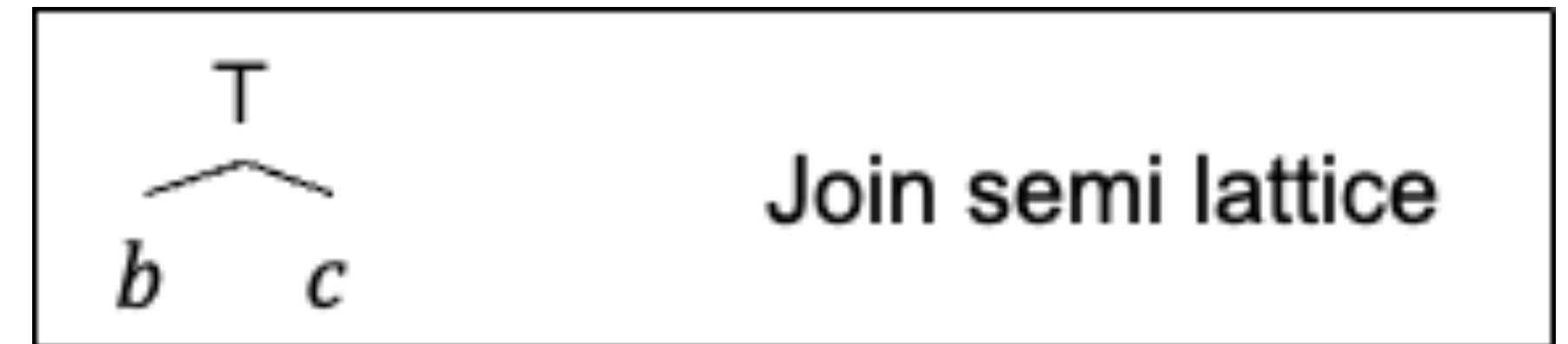
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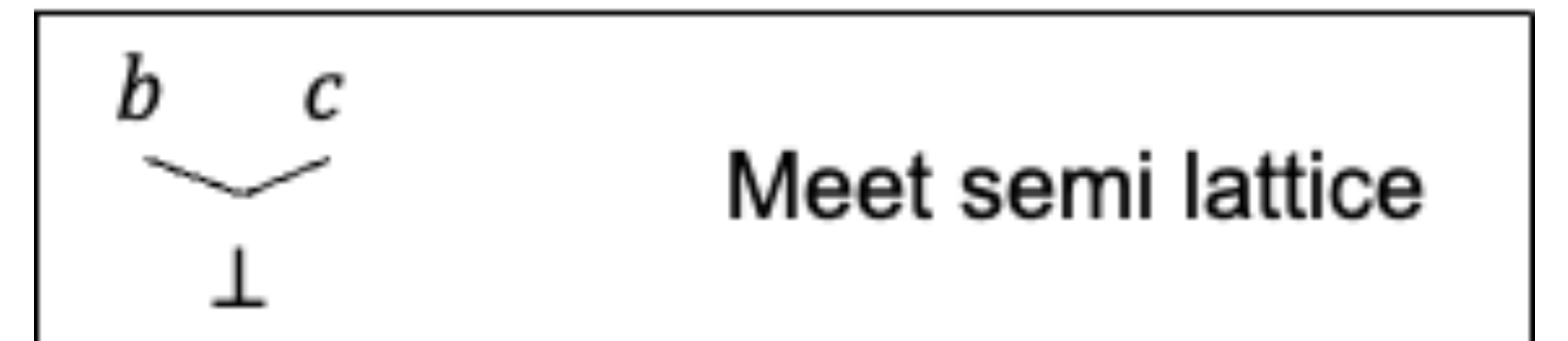
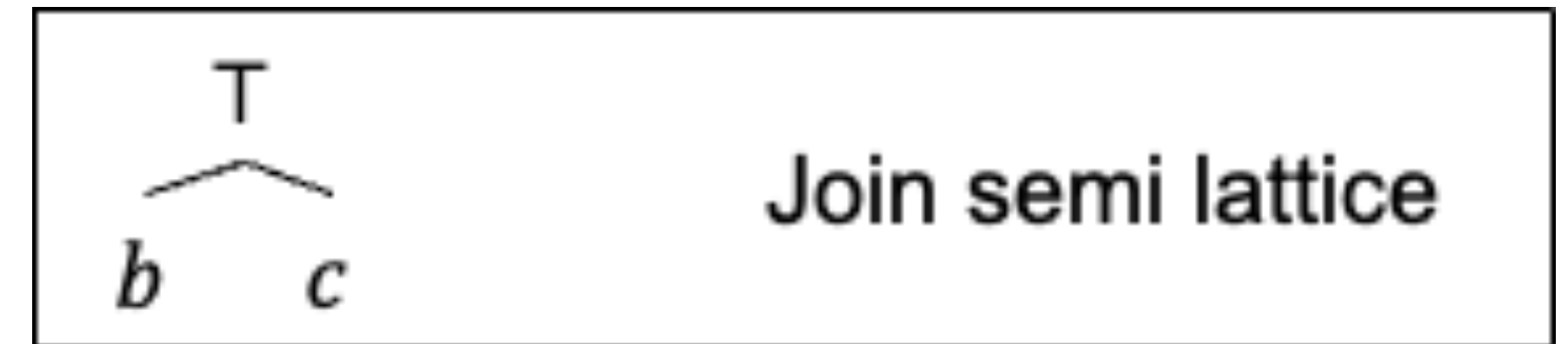
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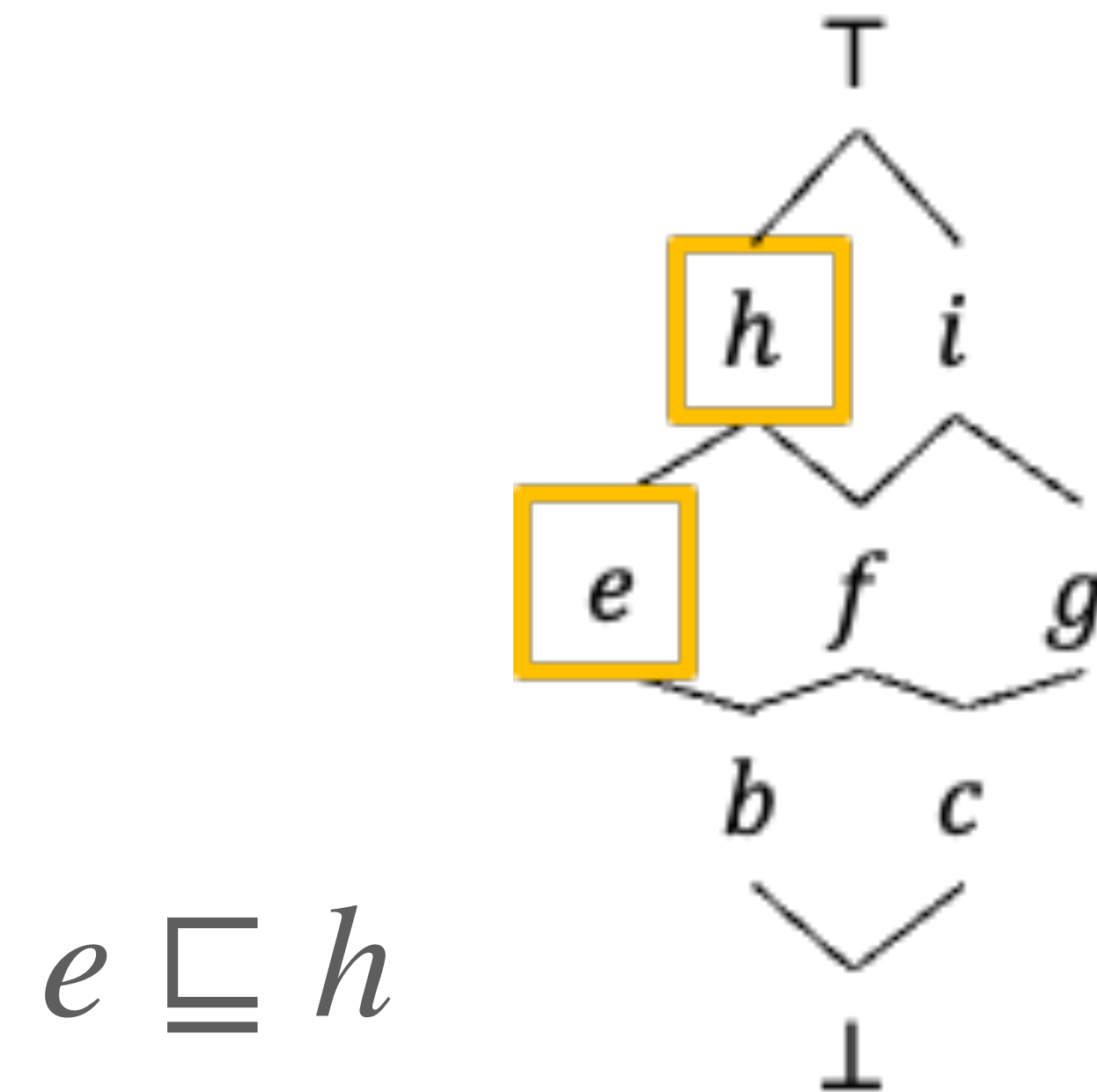
Lattice

- A poset $\langle X, \sqsubseteq \rangle$ is a lattice iff
 - $\forall x, y \in X. x \sqcup y$ exists (join semi lattice)
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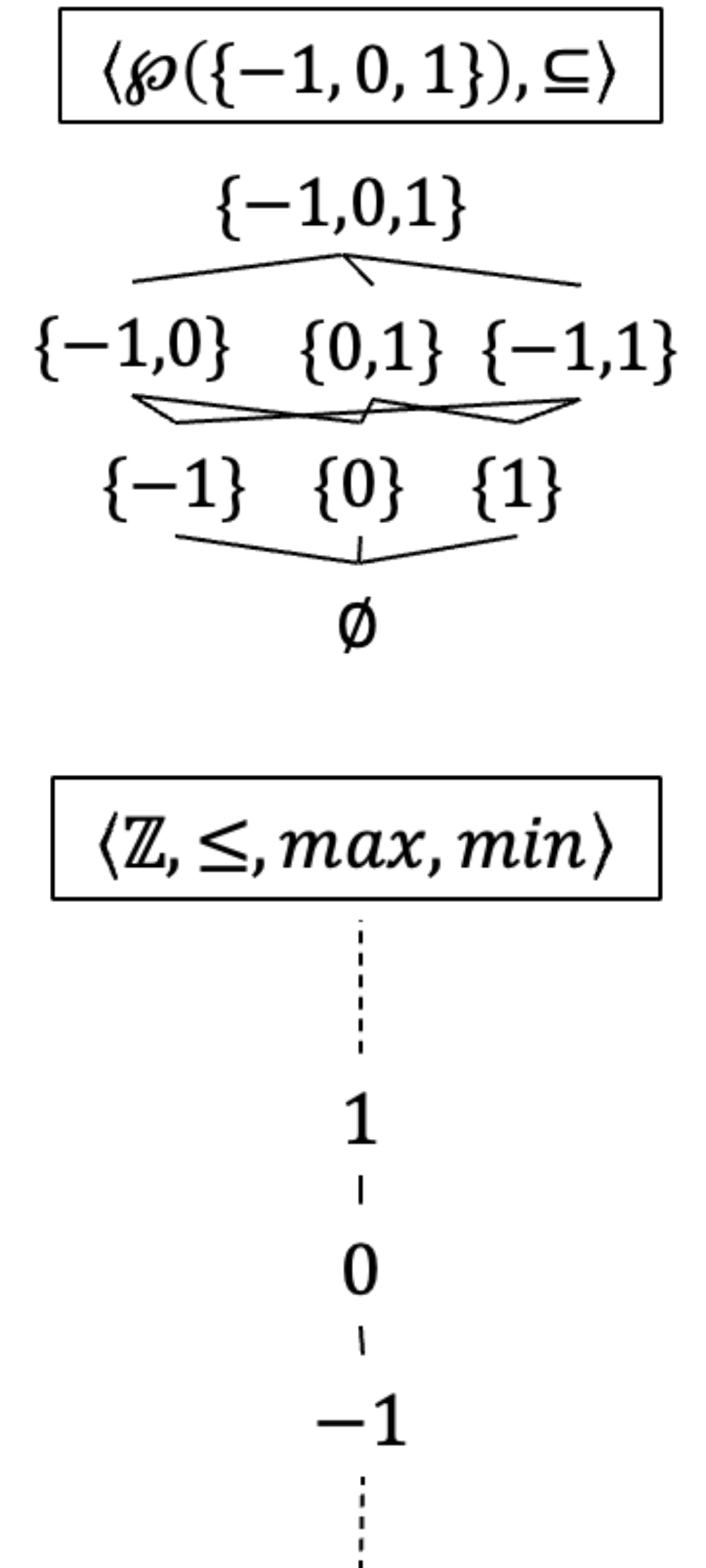
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$$\langle \mathbb{Z} \cup \{-\infty, +\infty\}, \leq, \max, \min \rangle$$

$$\begin{array}{c}
 +\infty \\
 \vdots \\
 1 \\
 | \\
 0 \\
 | \\
 -1 \\
 \vdots \\
 -\infty
 \end{array}$$

Complete lattice

Properties

Complete lattice

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- A complete lattice $\langle X, \sqsubseteq, \sqcup, \sqcap \rangle$ has bottom element $\perp$, a top element $\top$ and it is denoted by $\langle X, \sqsubseteq, \perp, \top, \sqcup, \sqcap \rangle$

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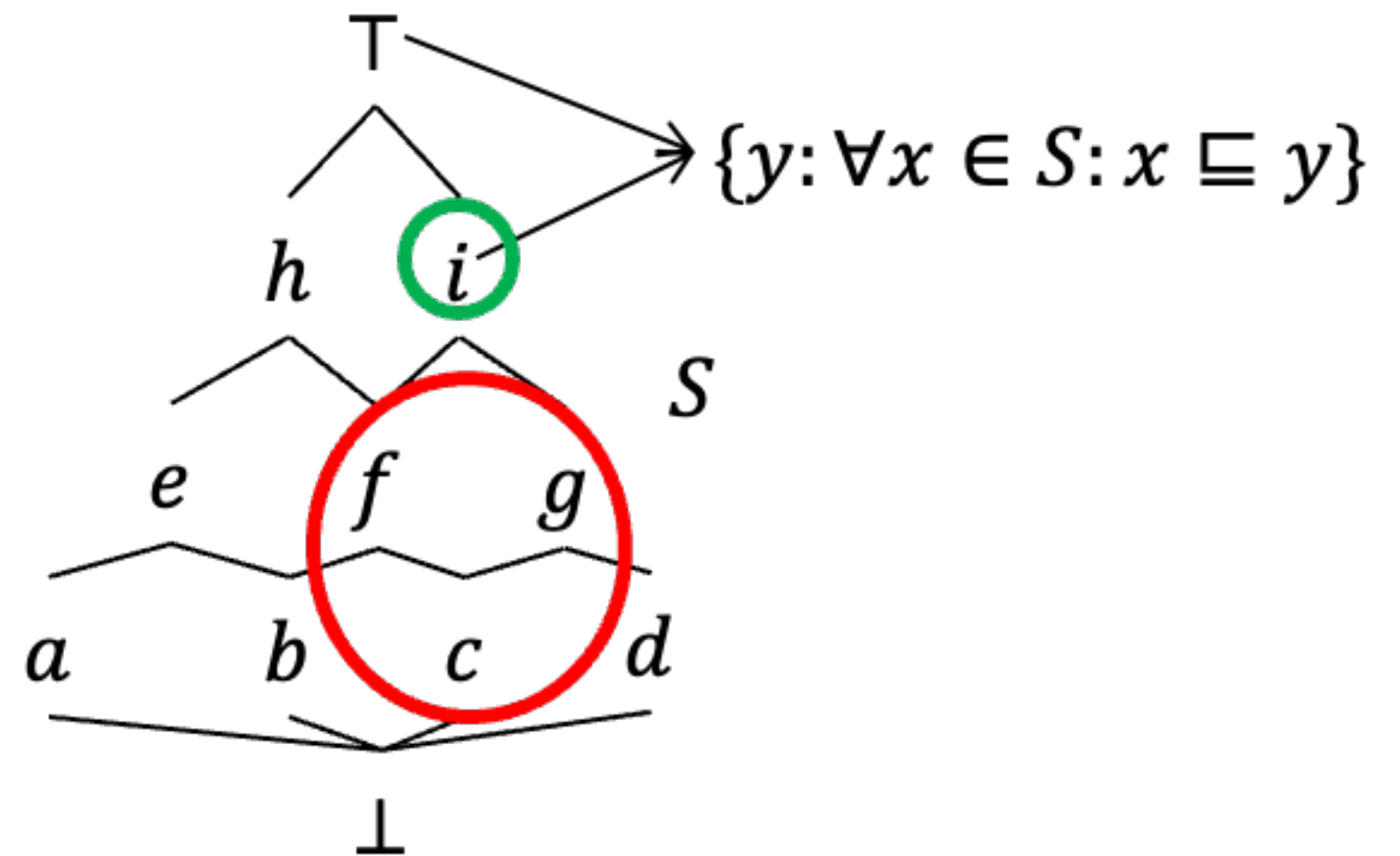
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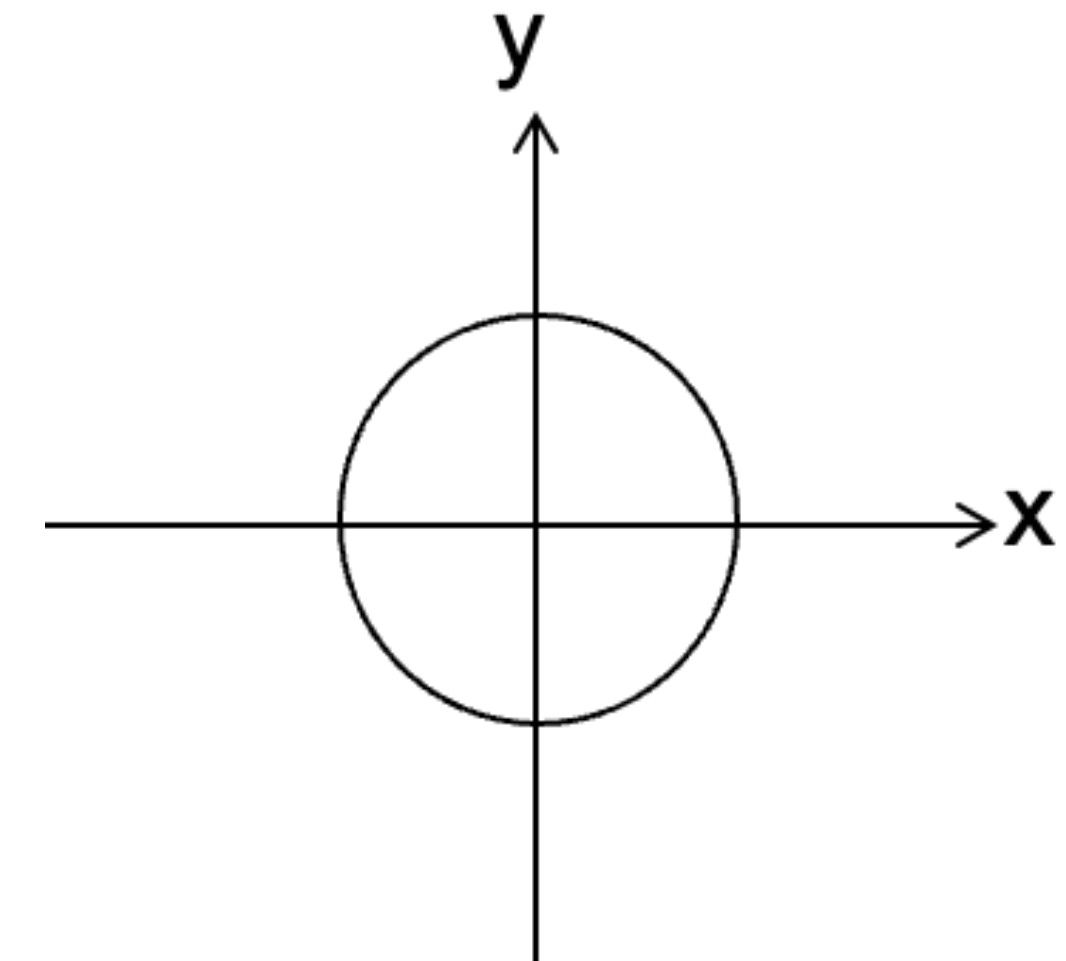
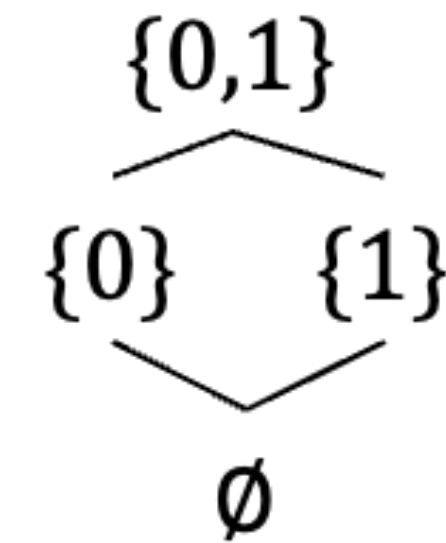
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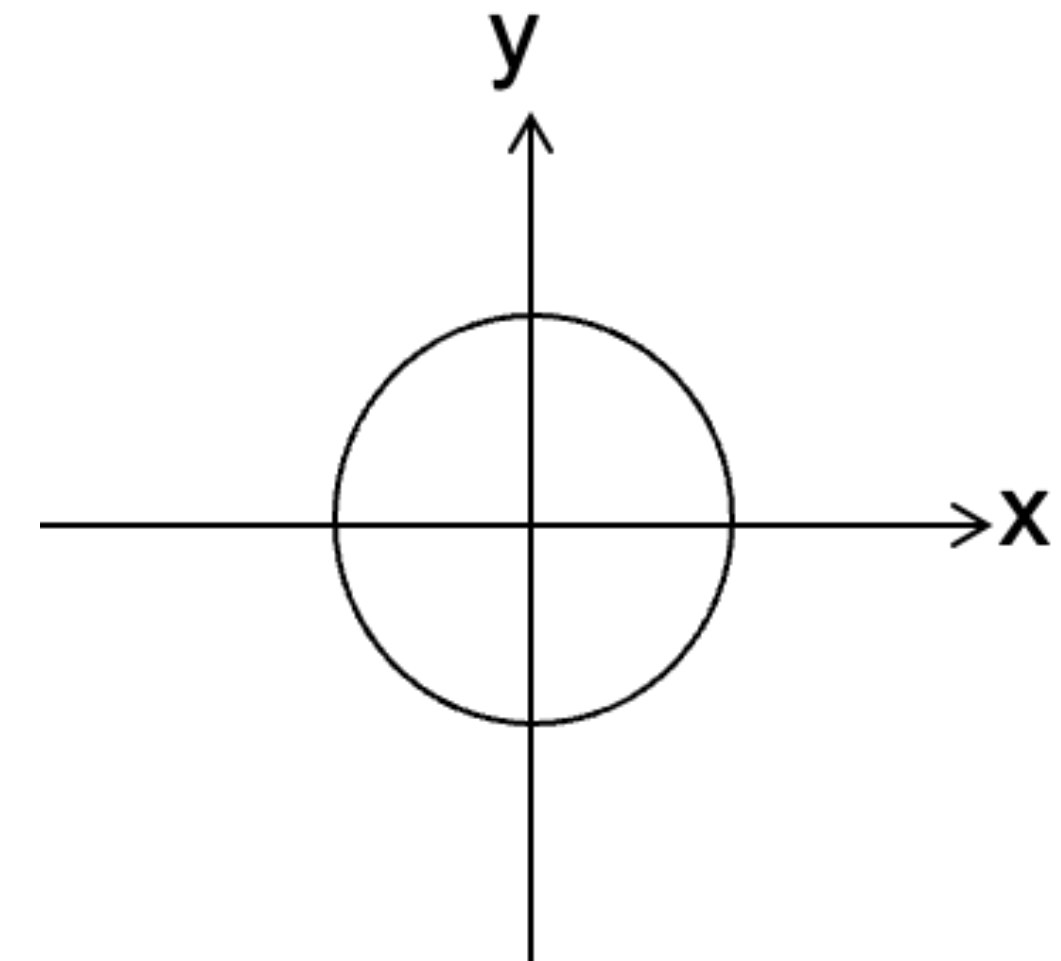
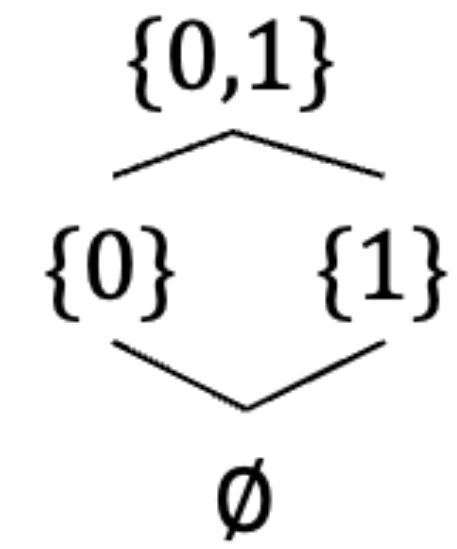
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$$\langle \wp(\{0, 1\}), \subseteq \rangle$$



Functions (or maps)

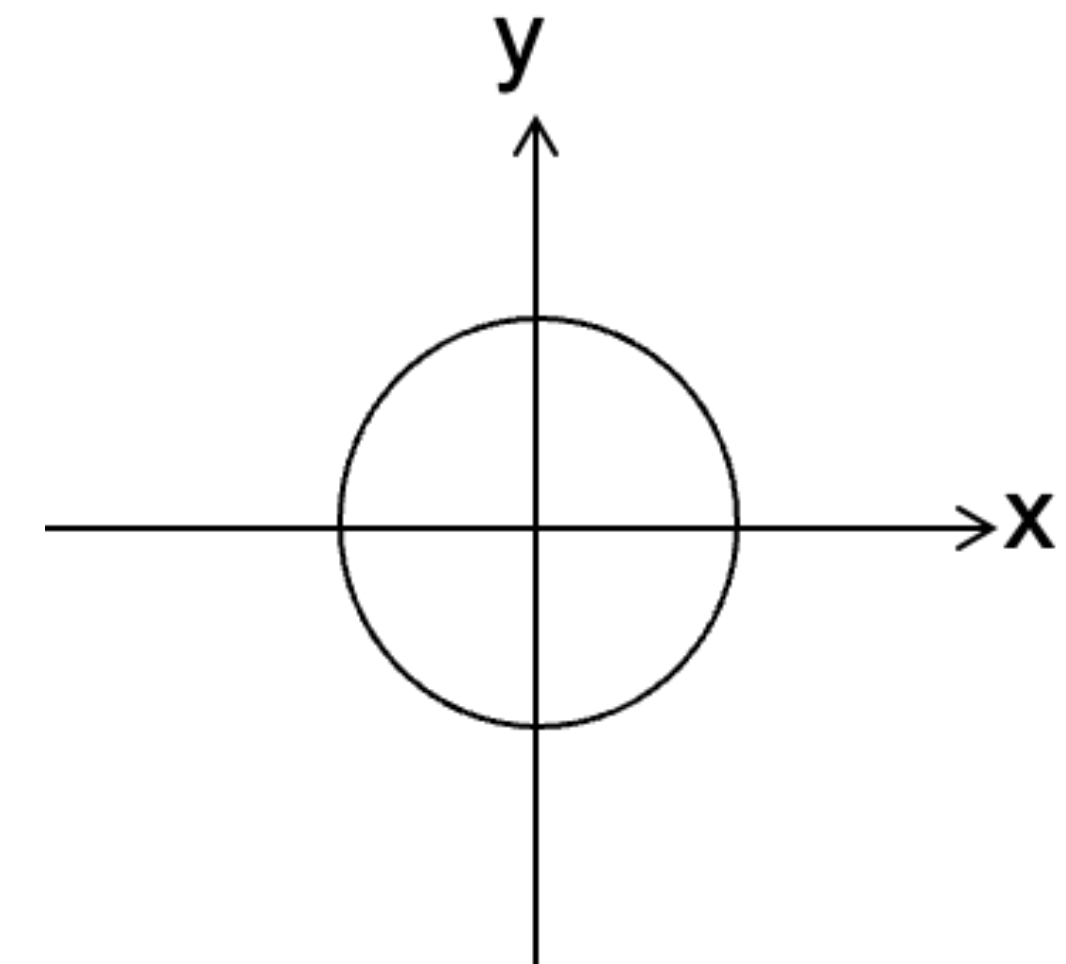
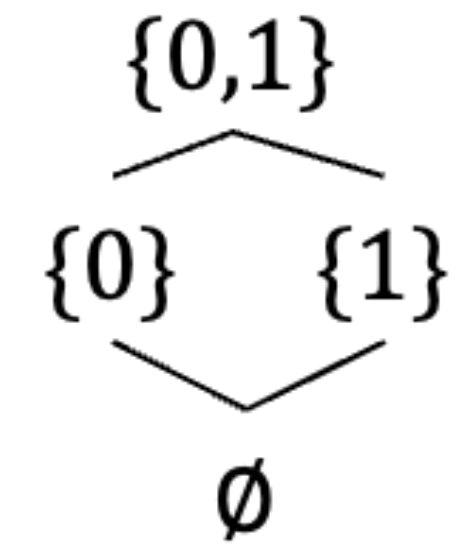
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Functions (or maps)

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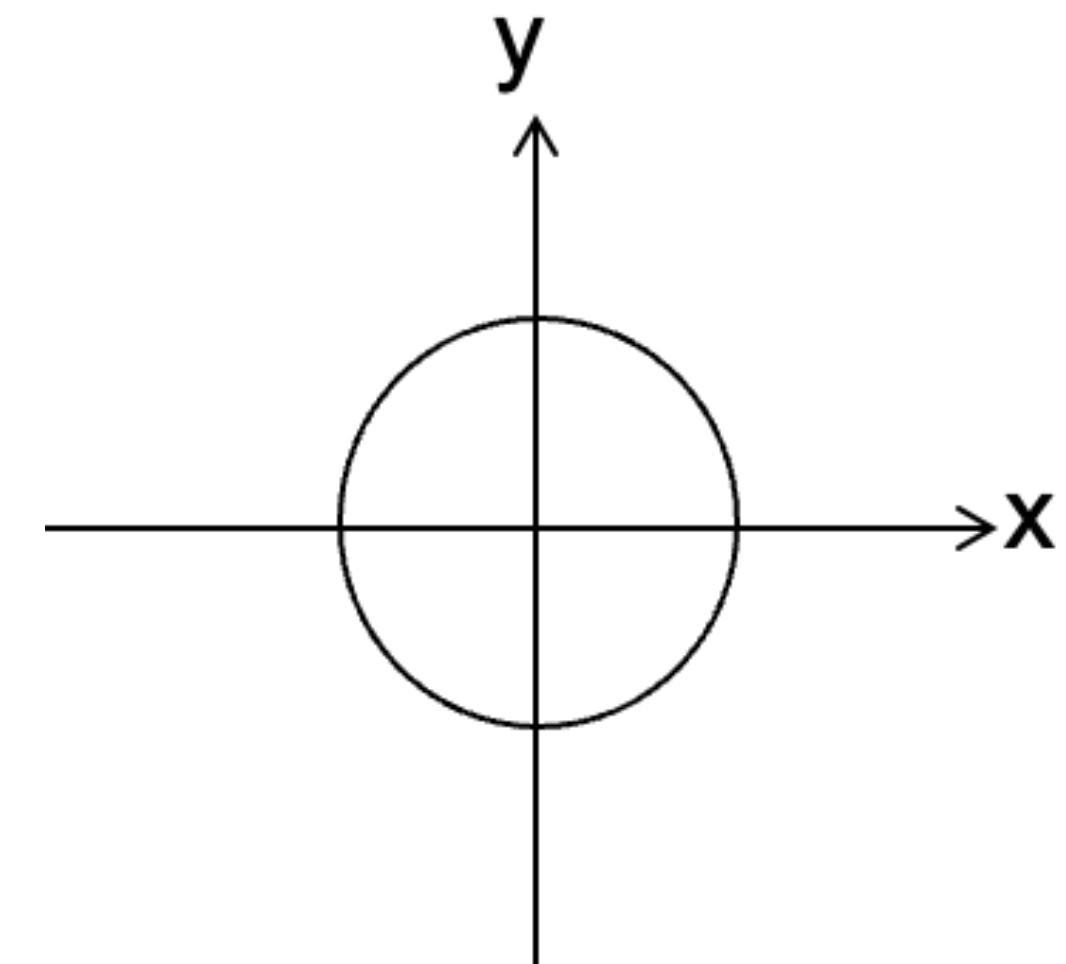
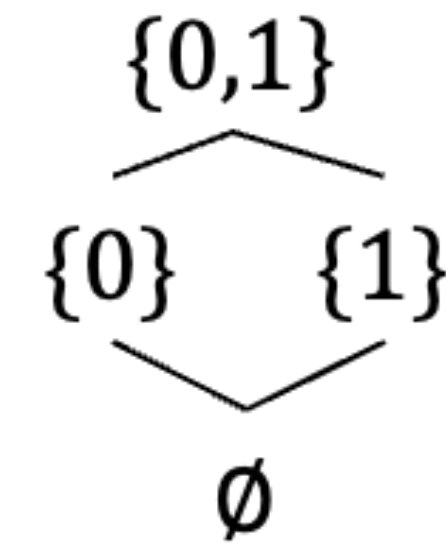
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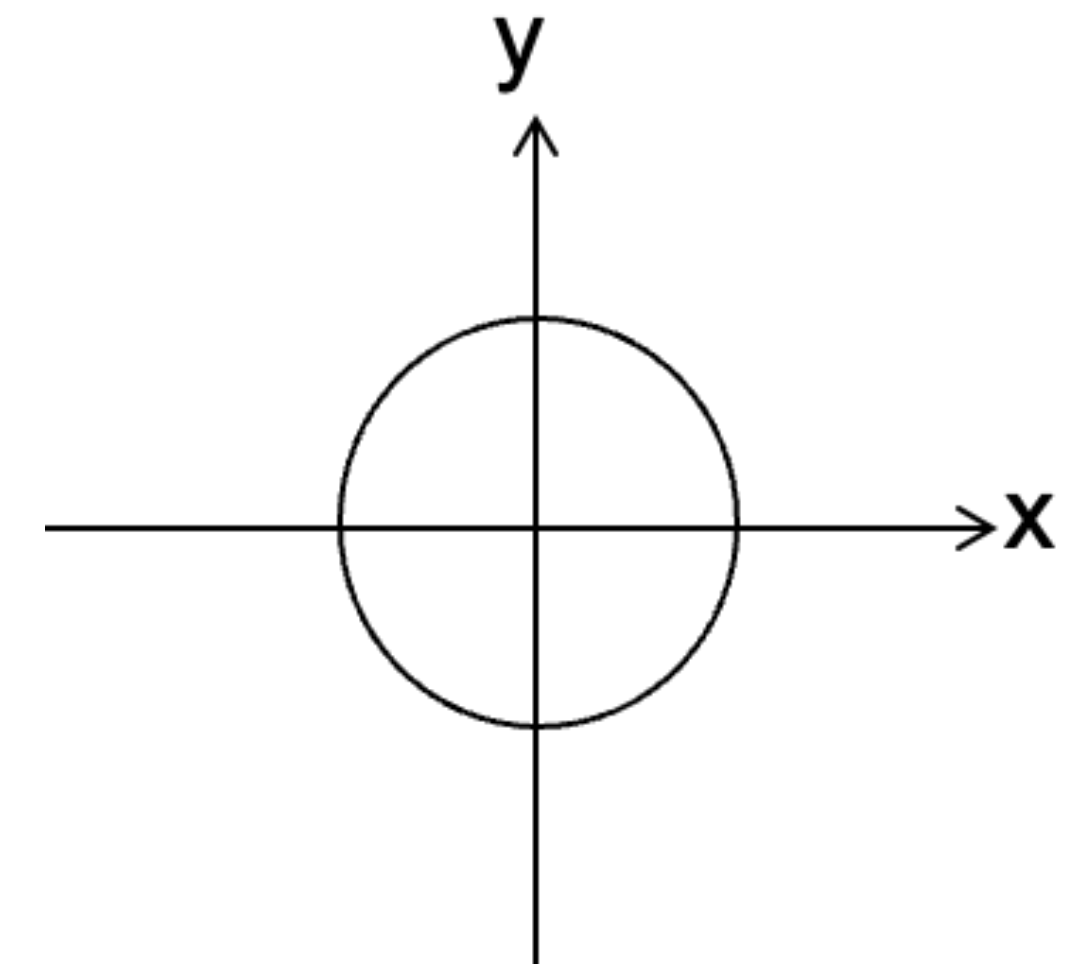
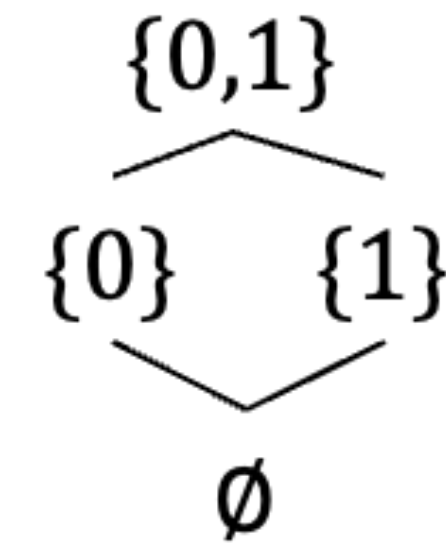
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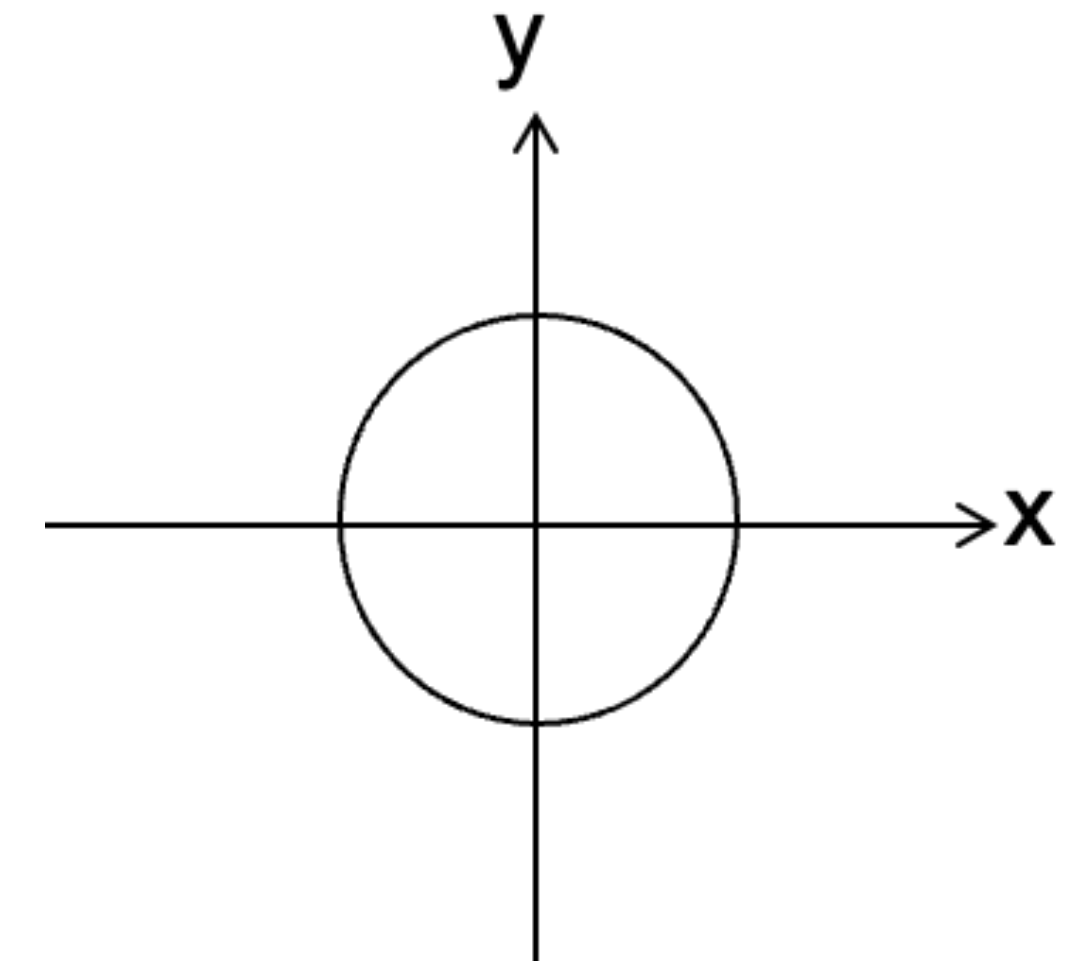
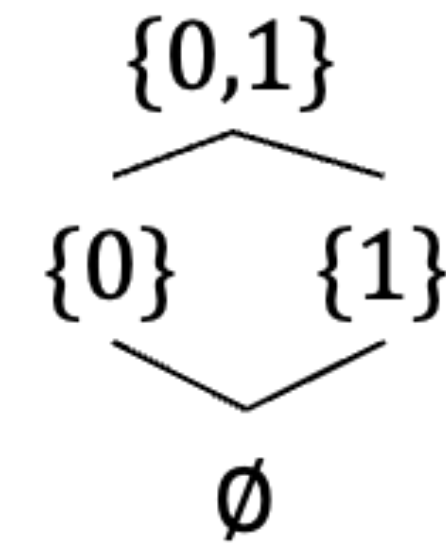
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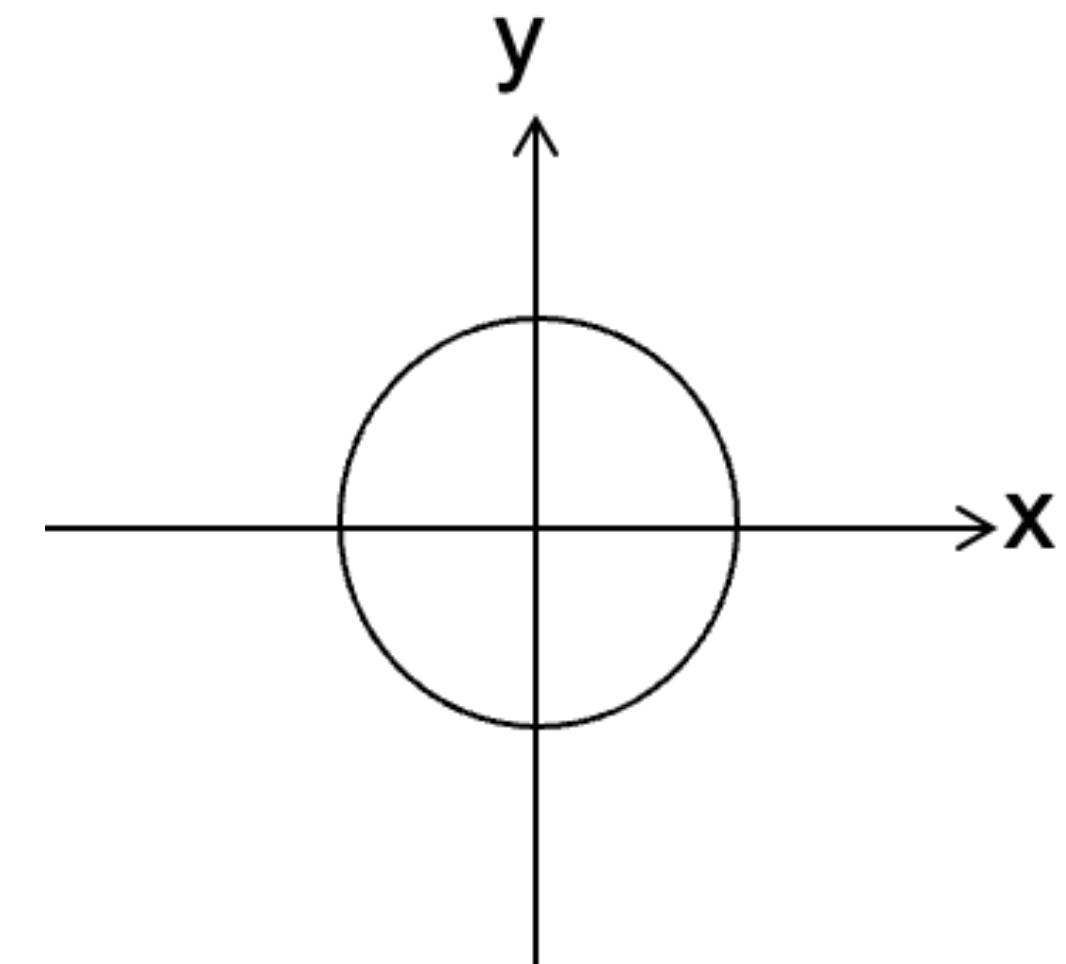
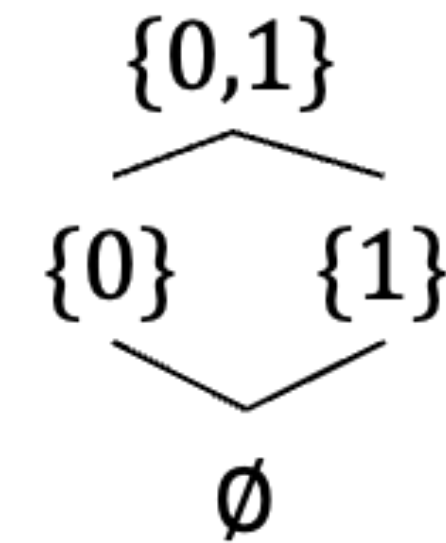
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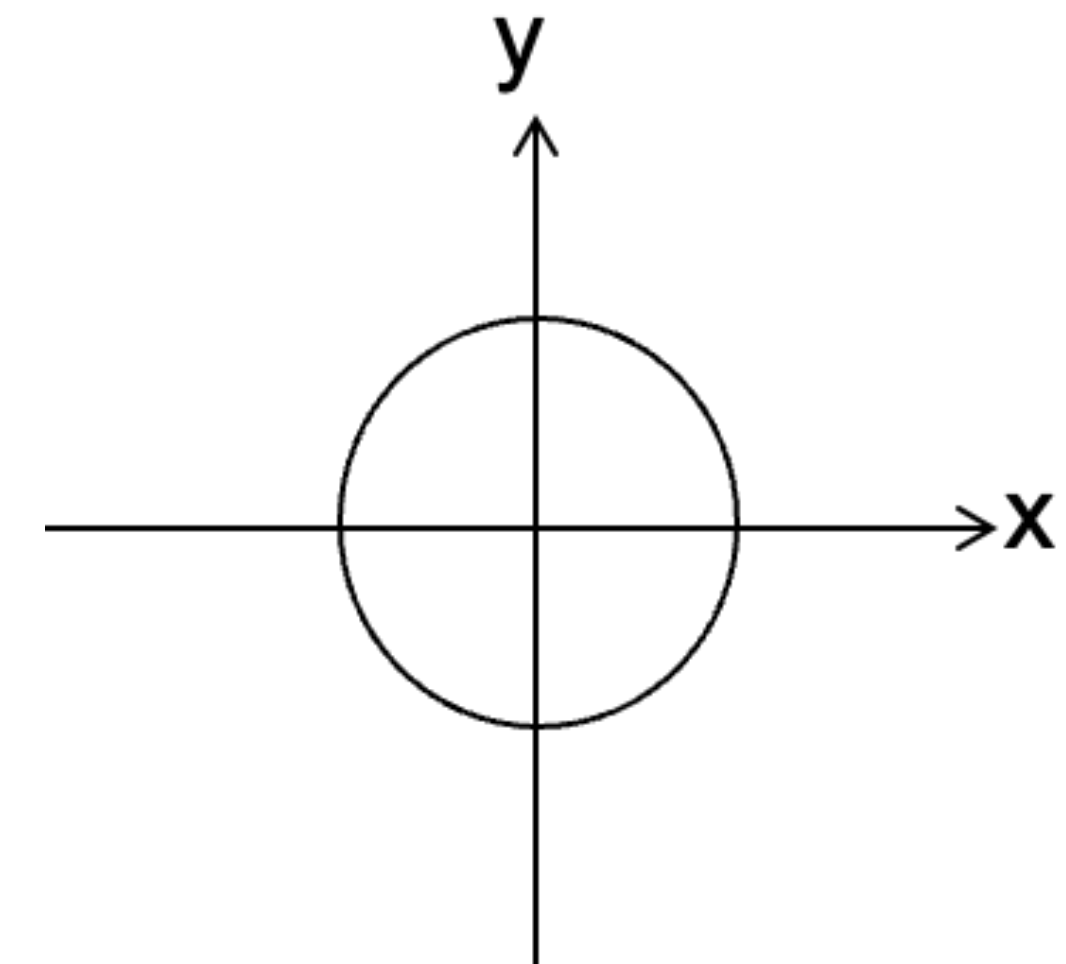
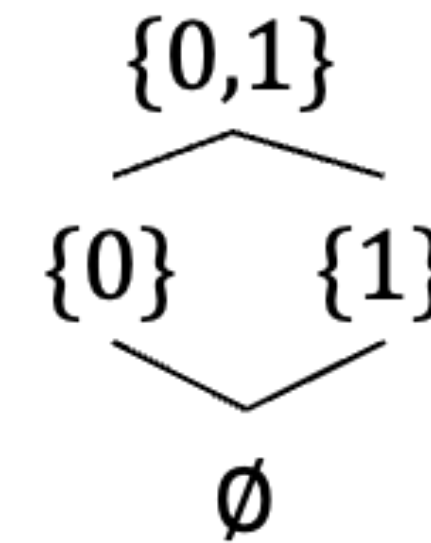
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$$f : \mathbb{Z} \rightarrow \mathbb{N}$$

$$f = [-1 \mapsto 1, 1 \mapsto 1] \\ = \{(-1, 1), (1, 1)\}$$

$$f' = f[2 \mapsto 2] \\ = \{(-1, 1), (1, 1), (2, 2)\}$$

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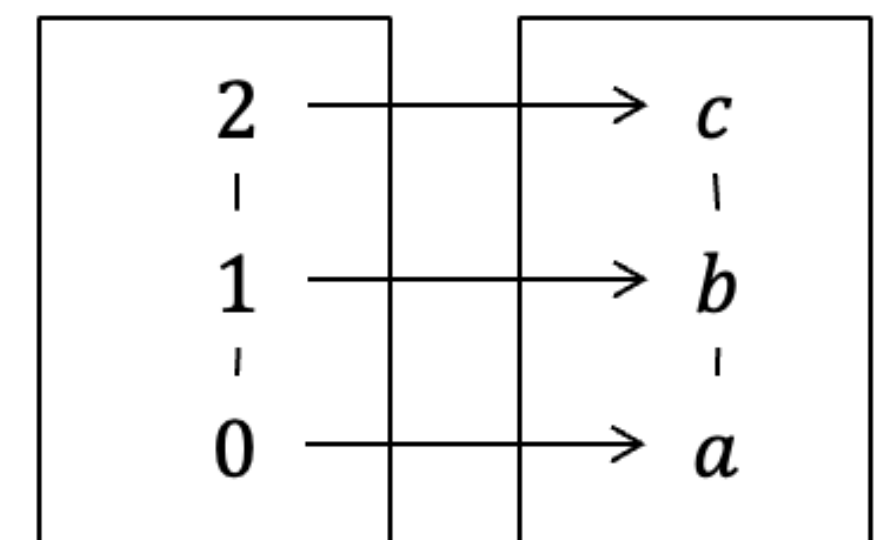
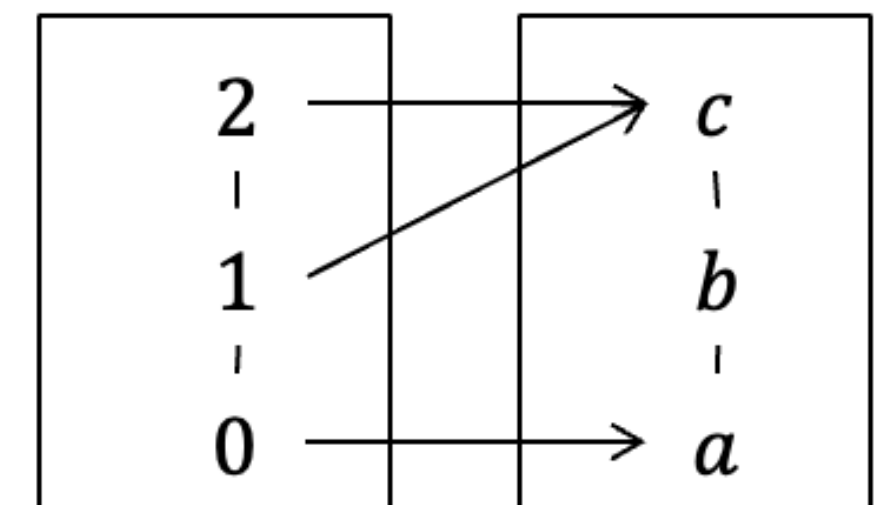
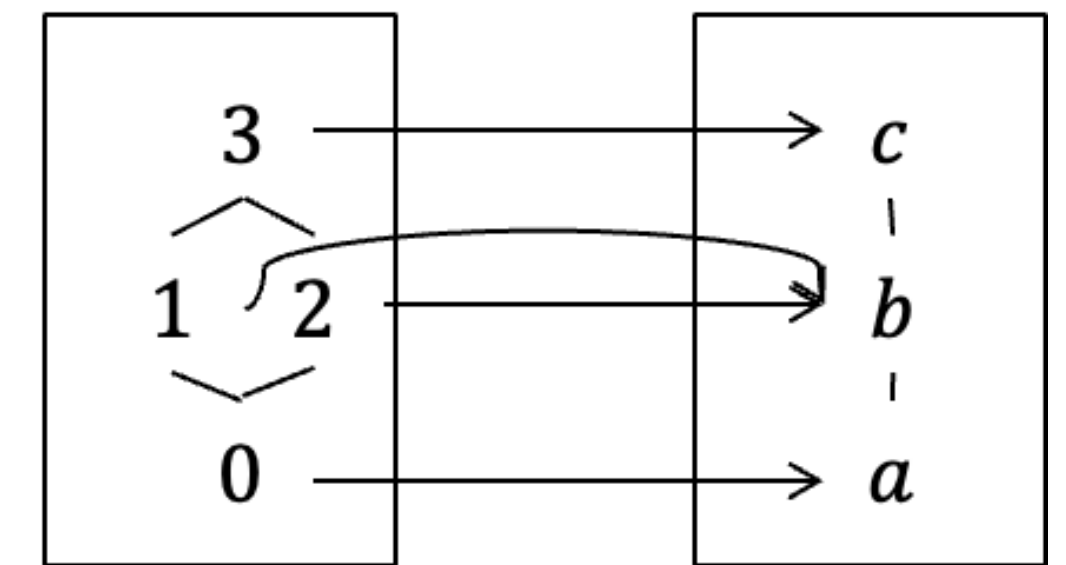
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 - surjective: $\forall y \in Y, \exists x \in X. f(x) = y$

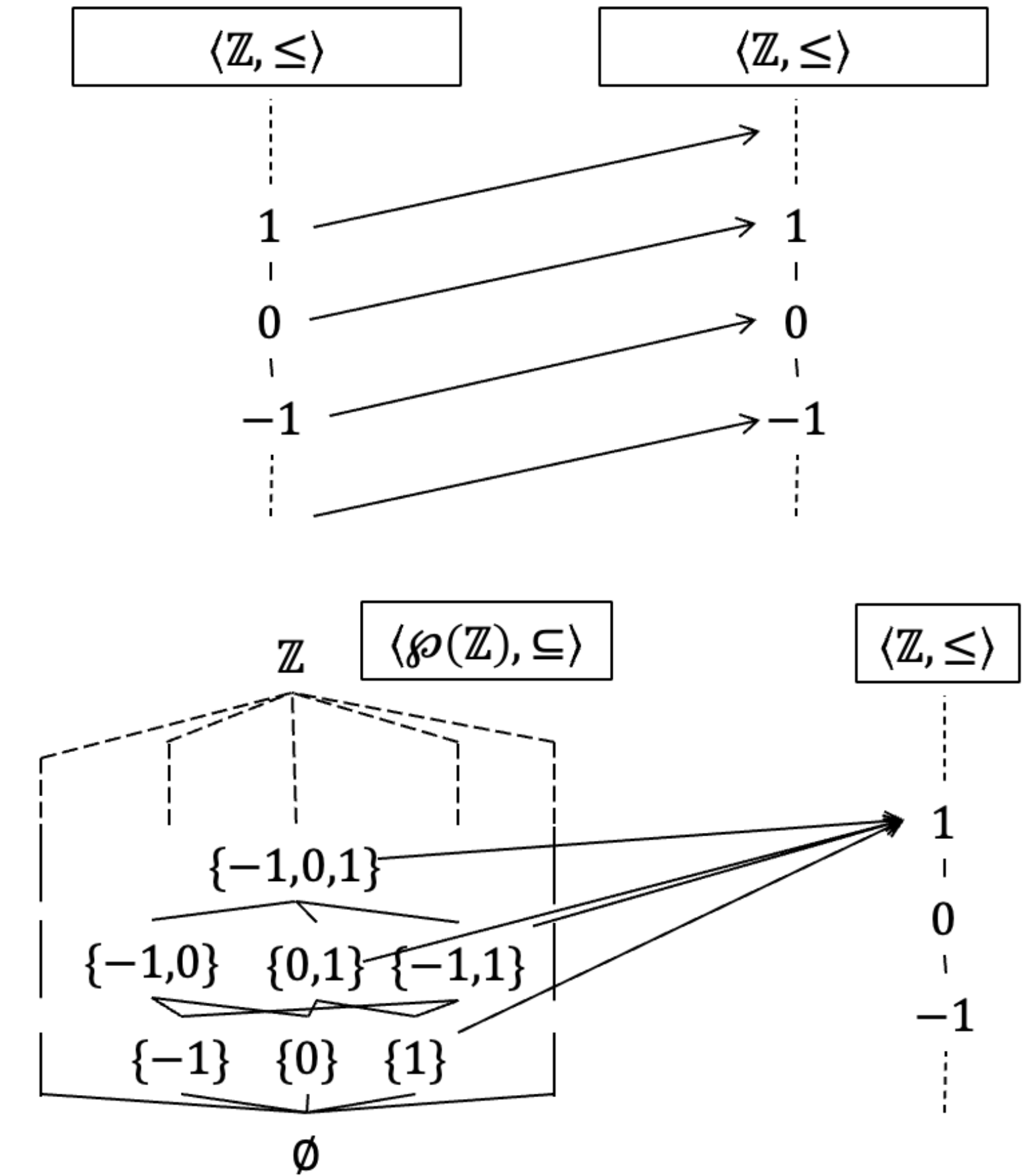
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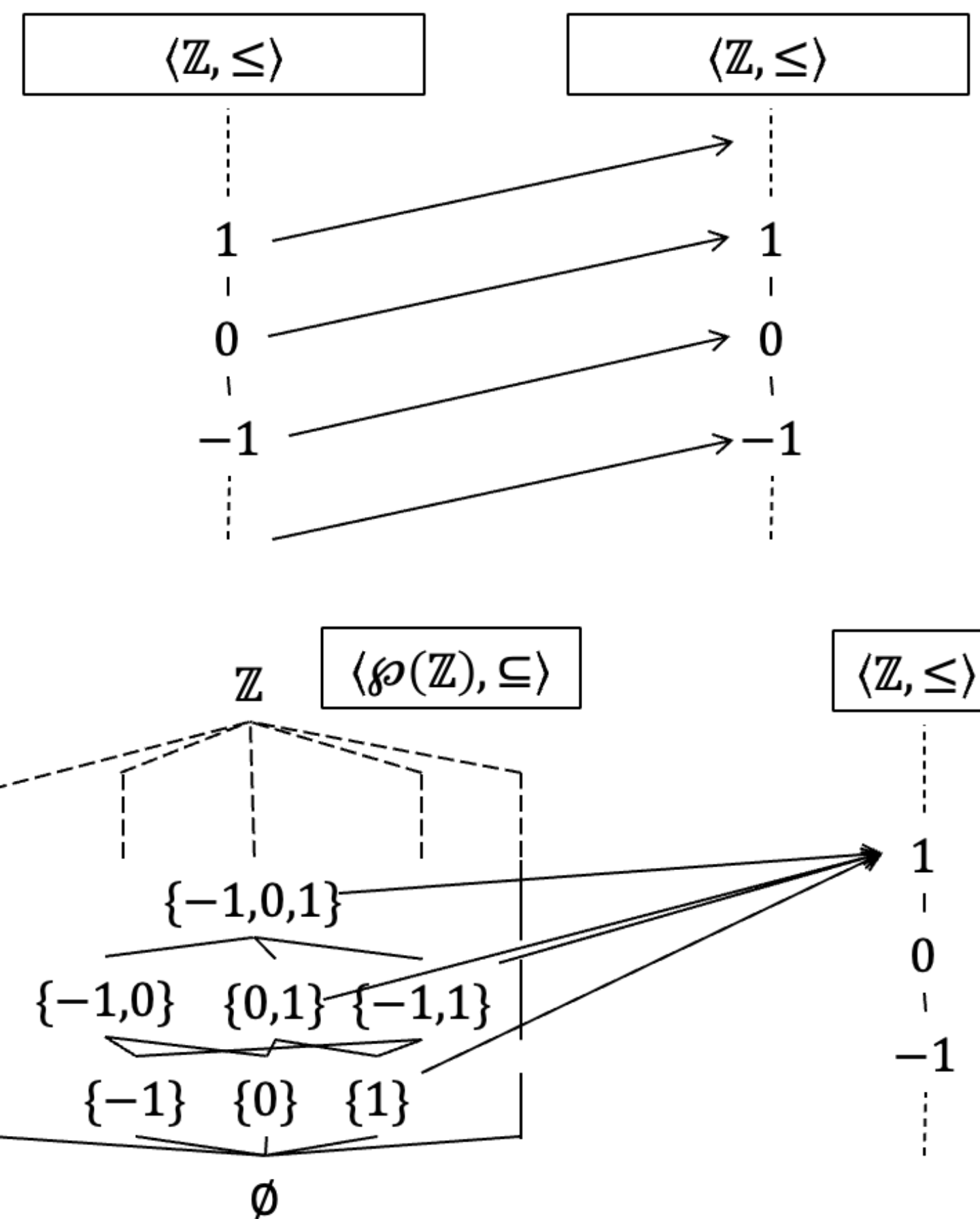
Example



Functions

Example

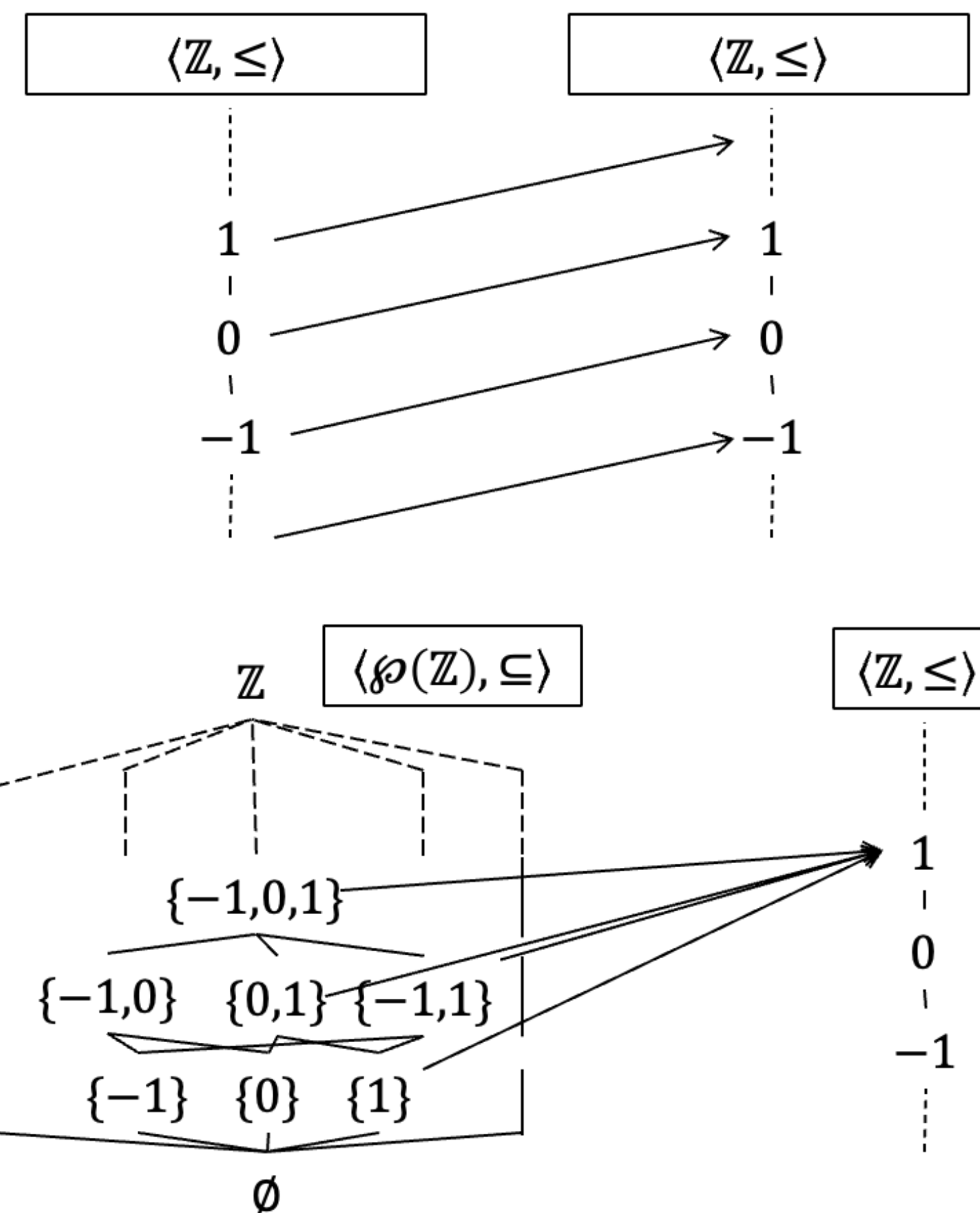
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Functions

Example

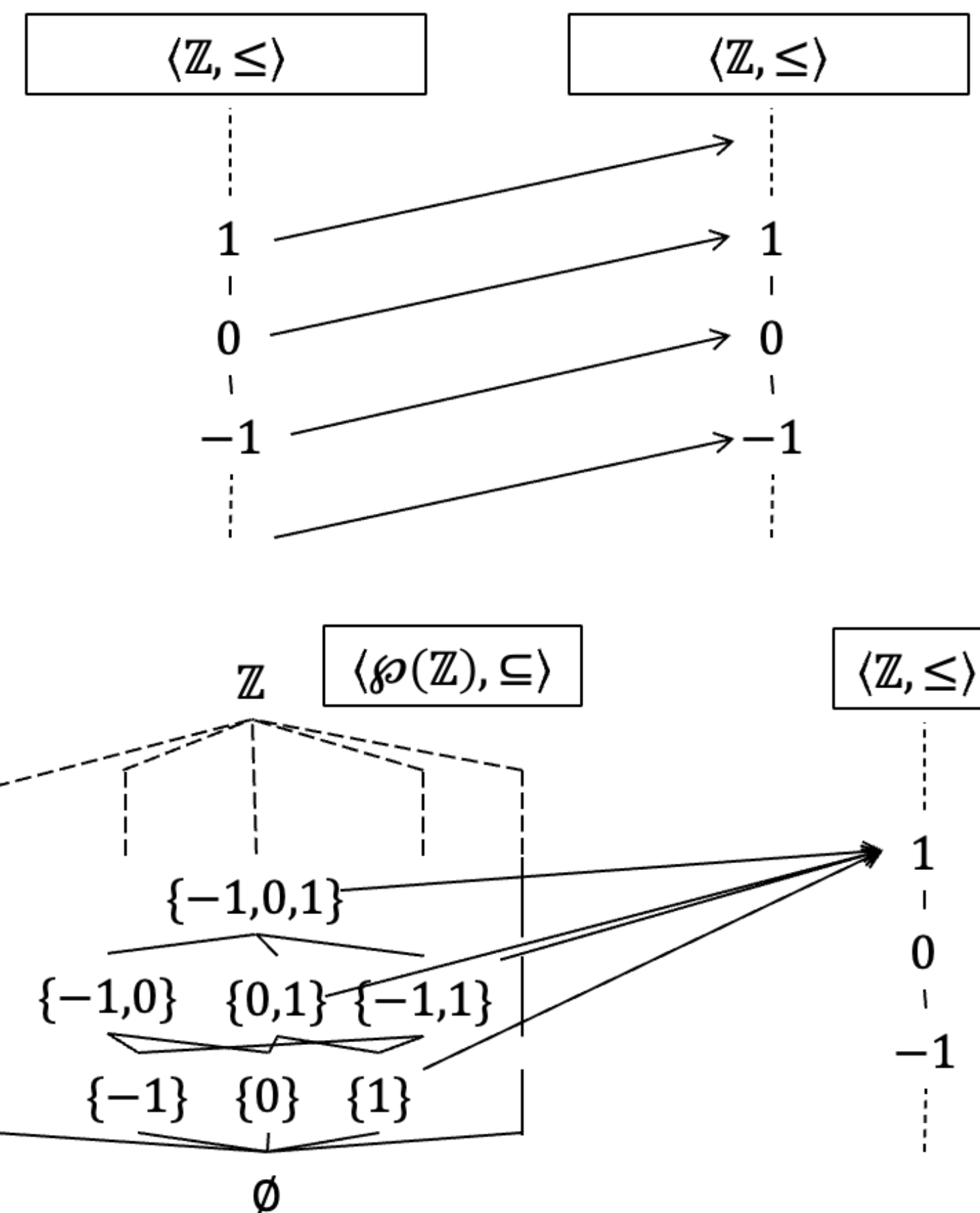
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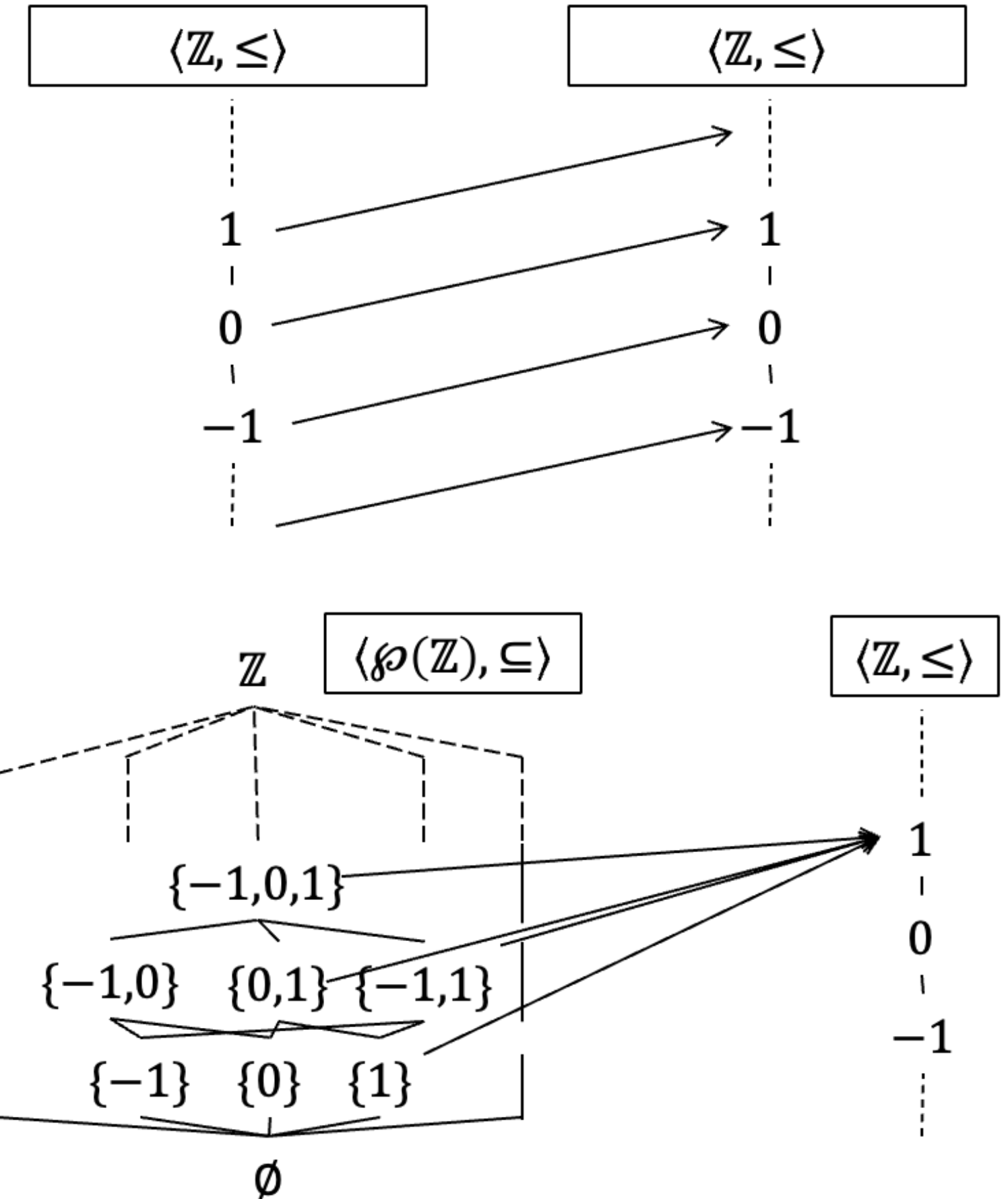
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Functions

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- $f: \langle \mathbb{Z}, \leq \rangle \rightarrow \langle \mathbb{Z}, \leq \rangle$
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- $f: \langle \wp(\mathbb{Z}), \subseteq \rangle \rightarrow \langle \mathbb{Z}, \leq \rangle$
 - $f(X) = \max(X)$, monotone, but not an embedding



Join/meet preserving functions

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- f is meet preserving if
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- A join or meet preserving function is monotone, but the opposite does not necessarily hold

Chains

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Chains

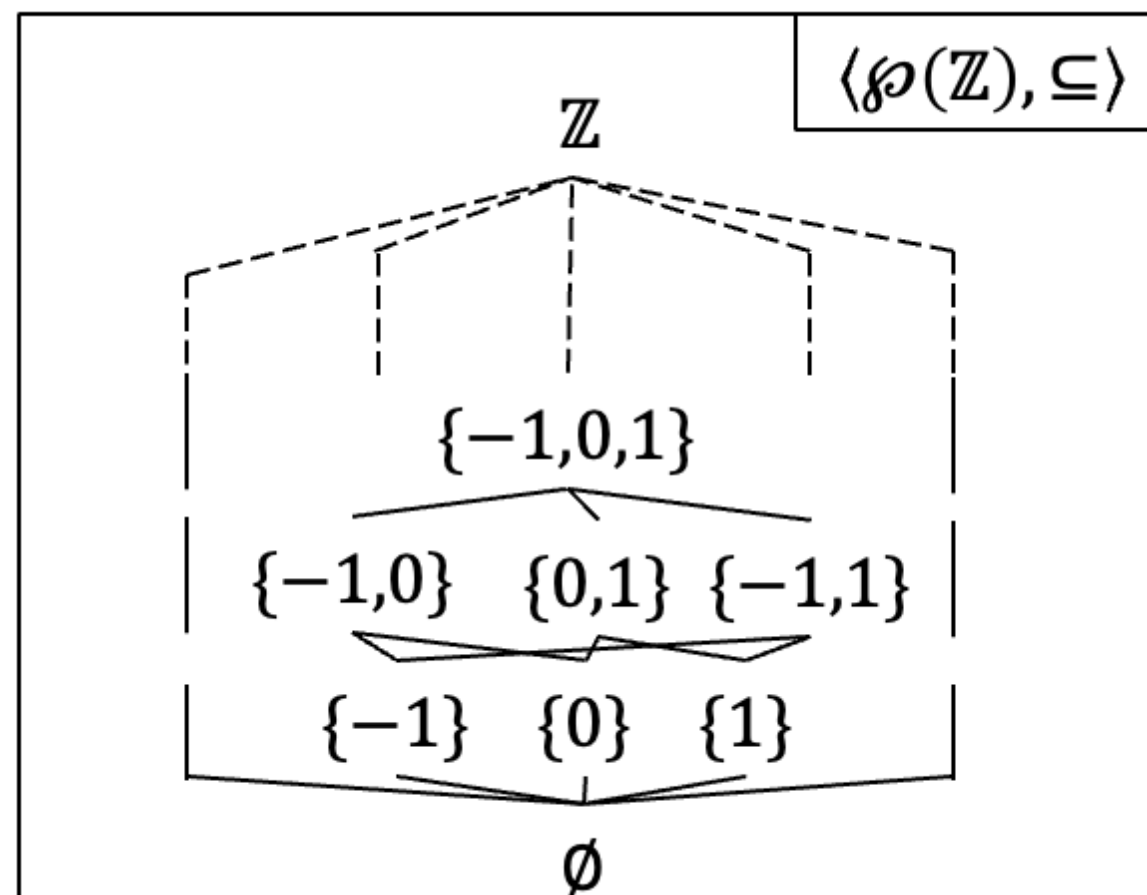
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- Examples of chains: $\{\emptyset\}$, $\{\{\emptyset\}, \{0,1\}\}$, $\{\{0\}, \{-1,0,1\}\}$

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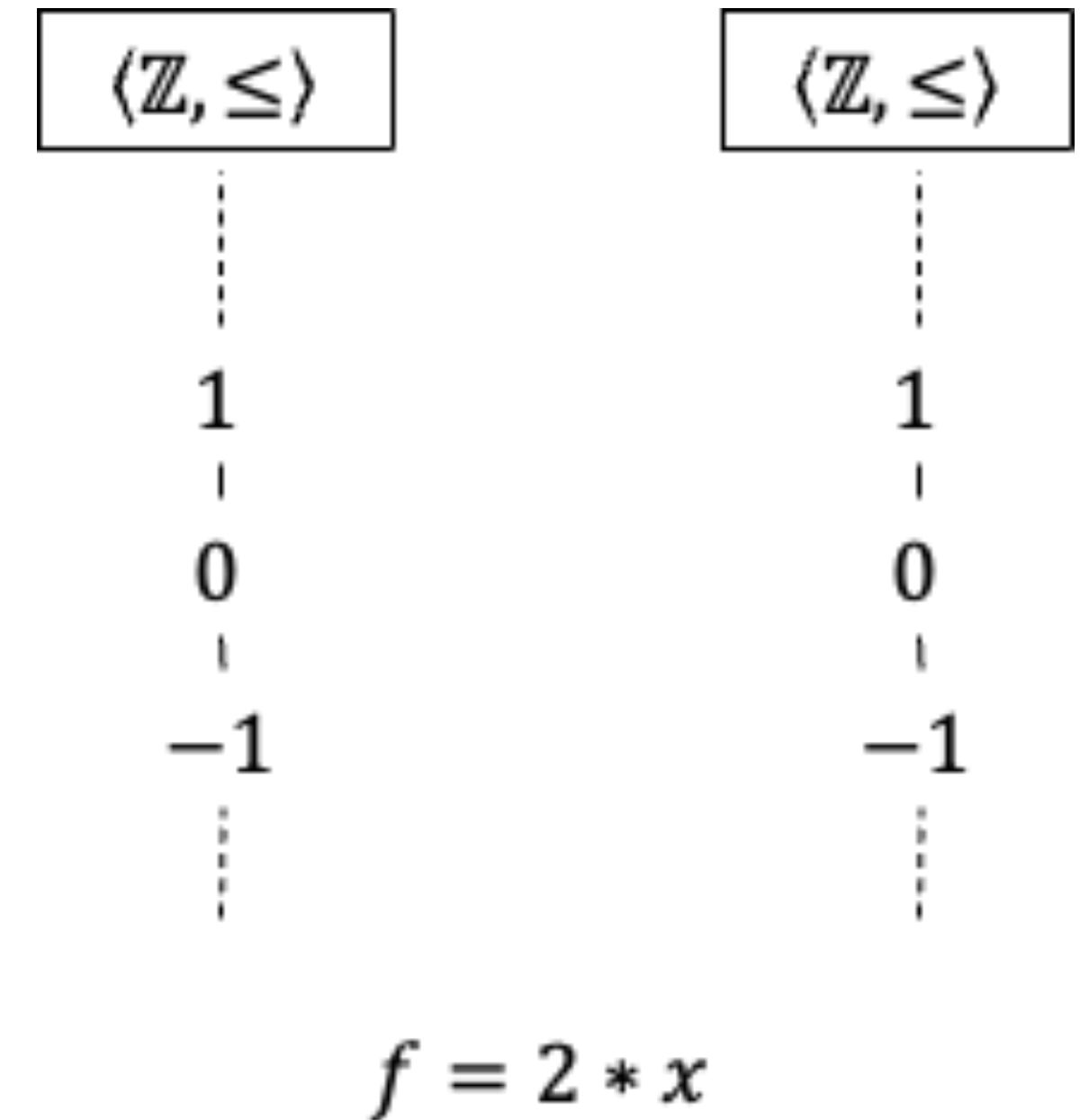
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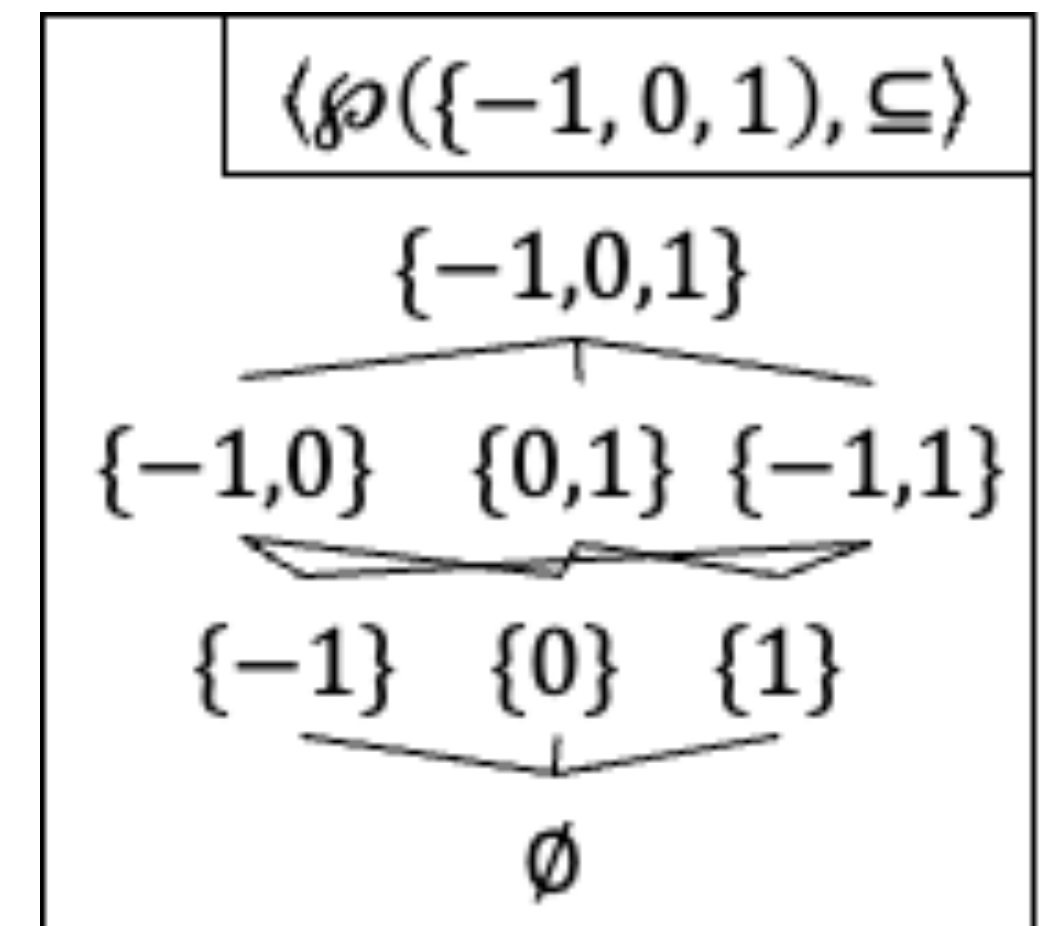
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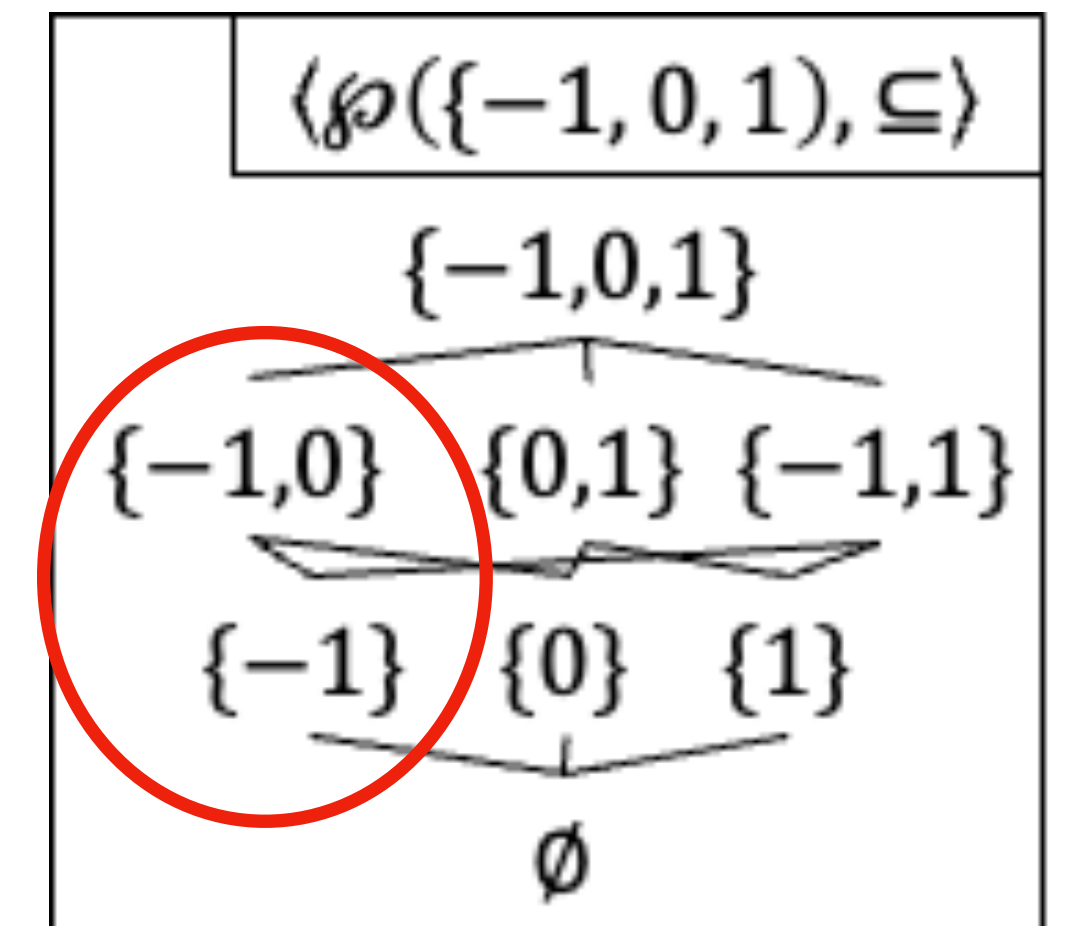
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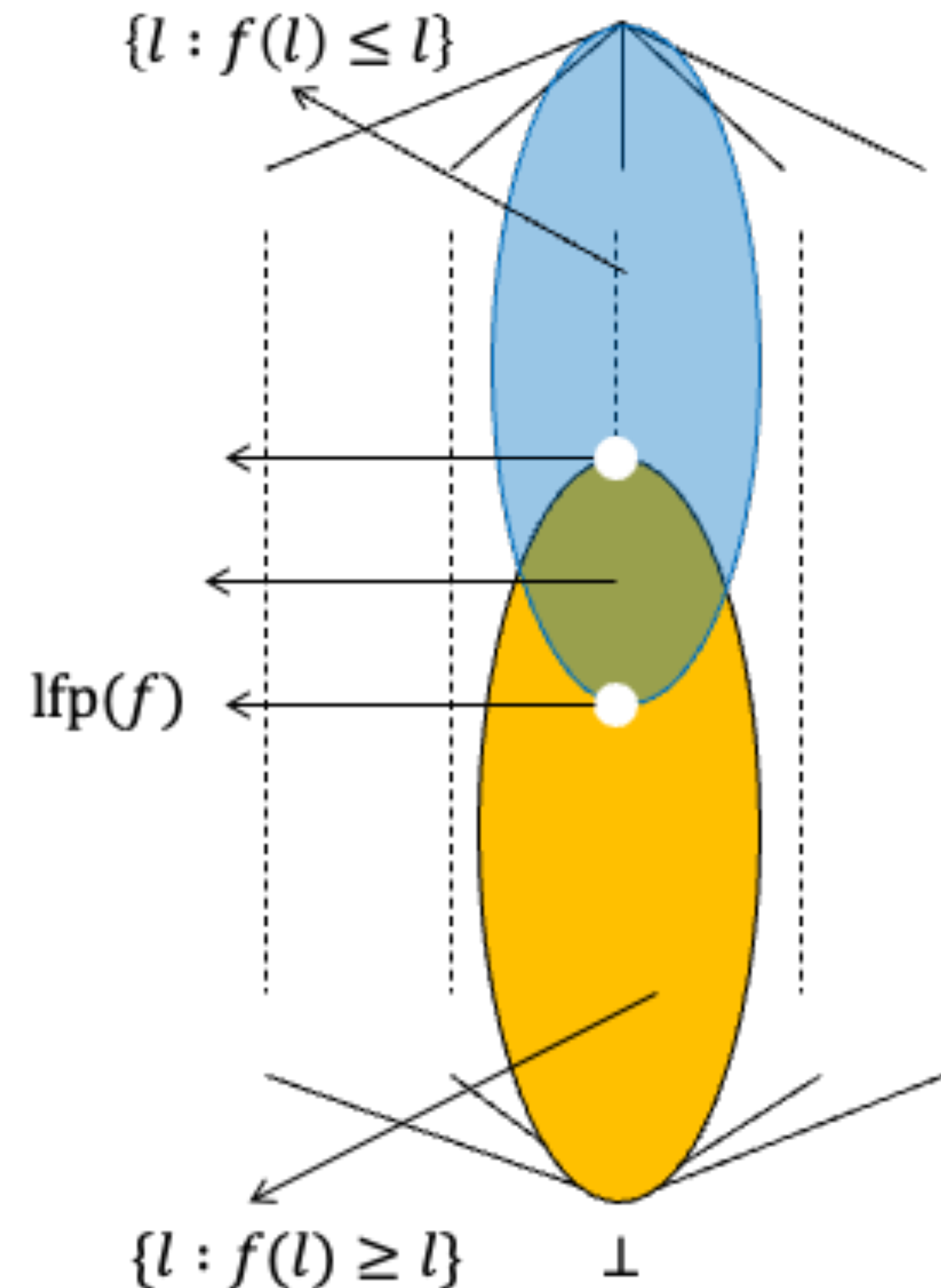
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Kleene fixpoint theorem

- Let $\langle X, \sqsubseteq, \perp, \top, \sqcup, \sqcap \rangle$ be a complete lattice, let $f: X \rightarrow X$ be a continuous function. f has a fixpoint and can be computed as $\sqcup_{n \geq 0} f^n(\perp)$